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Heat transport in confined nanostructures

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Abstract

Heat induced device failures limit the performance and speed of modern electronic devices. Once the device scales to the nanometre scale, managing and dissipating heat becomes a substantial task requiring an in-depth knowledge of transient thermal phenomena over nanometre scale. As modern electronic devices operate at GHz frequencies, nanosecond resolution is required to investigate the dynamic thermal phenomena. Obtaining both sufficient spatial and temporal resolution is still an experimental challenge. Optical nanothermometry techniques are spatially diffraction limited whereas contact probe techniques are temporally limited averaging the dynamic thermal phenomena. This work sets out to extend time-resolved scanning transmission electron microscope thermometry from a uniform thin film to a confined nanostructure, by following the thermal evolution along aluminium nanobridges 50 nm and 150 nm wide, patterned by electron-beam lithography on a 15 nm silicon nitride membrane. A focussed laser pulse heats the pad at one end of each bridge inside the electron microscope, and the temperature is measured from the shift in the aluminium bulk plasmon in time resolved electron energy-loss spectra. The measurement gives temperature maps at 59.4 nm spatial resolution and 2.08 ns temporal resolution. Heat appears first on the illuminated pad and about 30 ns later reaches the far end of the bridge in view. Further from the pad, along the bridge the peak rise of temperature is smaller, while the rise time and the delay to reach the peak temperature both grow; the two widths give similar behaviour. Finite-element simulations reproduce these trends. The method therefore reaches tens of nanometres and nanoseconds together on a structure well below the diffraction limit of optical techniques, opening new possibilities to study heat propagation in actual nanodevices. An example is given in thermoplasmonic nanopillars structure as a perspective.

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List of abbreviations and symbols

Abbreviations

ADF	Annular dark field
CFEG	Cold field-emission gun
CL	Cathodoluminescence
EBL	Electron-beam lithography
EELS	Electron energy-loss spectroscopy
FDTR	Frequency-domain thermoreflectance
FEM	Finite-element method
FWHM	Full width at half maximum
GDS	Graphic database system (layout format)
HAADF	High-angle annular dark field
IPA	Isopropyl alcohol
LPCVD	Low-pressure chemical vapour deposition
MFP	Mean free path
MIBK	Methyl isobutyl ketone
PCA	Principal component analysis
PEET	Plasmon-energy-expansion thermometry
PINEM	Photon-induced near-field electron microscopy
PMMA	Poly(methyl methacrylate)
ROI	Region of interest
SEM	Scanning electron microscope
SiN	Silicon nitride
SPIM	Spectrum image
STEM	Scanning transmission electron microscope
SThM	Scanning thermal microscopy
TBR	Thermal boundary resistance
ITR	Interfacial thermal resistance
TDTR	Time-domain thermoreflectance
TEM	Transmission electron microscope

ZLP	Zero-loss peak

Symbols

$\alpha(T)$	Linear thermal expansion coefficient
$\alpha_0, \alpha_1, \alpha_2$	Coefficients of the expansion polynomial for aluminium
ν	Poisson ratio
M, h	Biaxial modulus $E/(1-\nu)$ and layer thickness
c_p	Specific heat capacity at constant pressure
κ	Thermal conductivity, used interchangeably with k
σ, L_0	Electrical conductivity; Lorenz number
ρ	Mass density
ℓ	Bridge length
w	Bridge width
$E_p(T)$	Bulk plasmon energy
ΔE_p	Plasmon energy shift, $E_p(T_0) - E_p(T)$
$\hbar\omega_p$	Plasmon quantum
$n(T)$	Free-electron density
m, e	Electron mass and charge
ε_0	Vacuum permittivity
λ	Laser wavelength, 550 nm
Λ	Carrier mean free path
Q	Volumetric heat source
q	Heat flux
R	Thermal boundary resistance
G	Electron–phonon coupling constant (sec. 1.2)
T_0	Reference temperature of the plasmon calibration
ΔT, $\Delta T_{\max}$	Temperature rise and its peak value; also the temperature step across an interface (sec. 1.5)
τ_{rise}	Rise time constant
τ_{decay}	Decay time constant
$t_{\text{on}}, t_{\text{off}}$	Laser turn-on and turn-off
t_{peak}	Time of the temperature maximum
$t_{\text{rise}\to\text{peak}}$	$t_{\text{peak}} - t_{\text{on}}$
A, C	Transient amplitude and baseline; c is also the volumetric heat capacity
s_x, s_y	Scan coordinates

Introduction

Heat induced damage is a major cause of failure in modern nanoscale electronic devices [1]. As the device size shrinks, thermal failures limit their performance and lifetime. This is because at these scales heat is carried by phonons and electrons whose behaviour differs from the bulk. Due to enhanced surface-to-volume ratio, scattering at surfaces and grain boundaries degrade the thermal conductivity, causing local thermal hotspots [2, 3, 4]. In a nanodevice these local hotspots affect the electrical properties, leading to local heat induced damage.

In a quantum device, local thermal effects are one of the major causes of decoherences. Designing an efficient thermal management system at a nanometer scales become highly important for such a device [5].

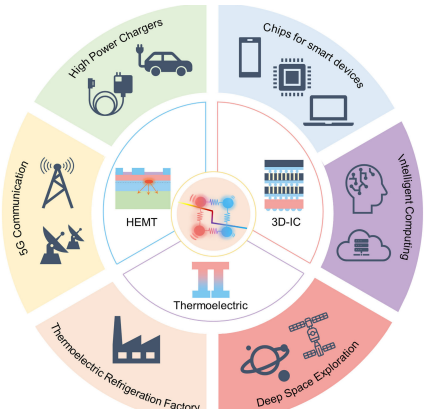


FIGURE 1: Scope of thermal management and phonon engineering at the nanoscale for advancements of modern devices. Reproduced from Xu *et al.* [6], figure 1.

Designing efficient devices at a nanoscale (fig. 1) require an in-depth understanding of the heat transport not only in an equilibrium but also during the thermal transient and non-equilibrium phase. Along with the temporal information, understanding the local thermal hotspots and physics behind them will help engineers design efficient thermal management systems, boosting the chip performance.

Various techniques exist to map the temperature profiles of a nanodevice, measuring their thermal properties. The existing nanothermometry techniques including Raman thermometry, thermoreflectance and scanning thermal microscopy (SThM) are advantageous, suited to particular materials and have a limited scope of appli-

cation. Optical techinques offer a high temporal resolution but are limited by the diffraction limit in spatial resolution. SThM offer a high spatial resolution but being a local contact probe technique, introduce thermal contact resistance and is limited by the temporal resolution.

In the STEM group at the Laboratoire de Physique des Solides, Orsay where this intership takes place, a new approach has been recently developed using a scanning transmission electron microscope (STEM) to access such information. F.Castioni *et al.*[7] performed nanothermometry on uniform aluminium thin films, using the electron energy loss spectrum to map out the local temperature of the thin film with a temporal resolution of $\sim 1ns$ [8] and a high spatial resolution offered by a STEM. This work applies the principle of electron energy loss spectroscopy to investigate the heat flow dynamics and explores the effect of size contraints on heat dissipation. We use a model system of aluminium nanobridges to establish the technique and understand the physics behind the heat transport by comparing the experimental results with finite element simulations performed in COMSOL.

This work is divided into six chapters. Chapter 1 presents the theoretical background on thermal transport and compares the existing nanothermometry techniques. Chapter 2 provides the theoretical and practical knowledge of electron microscopes, electron energy loss spectroscopy and explains the modified STEM - ChromaTEM used for this work. Chapter 3 focusses on the finite element simulations performed in COMSOL, the nanobridge designs, the model system used in this work and provides a detailed catalogue of the fabrication process developed as a part of this internship. Chapter 4 focusses on the data acquisition and data structure of the recorded data and provides the algorithms and workflows developed for data processing and data analysis. Chapter 5 reports the temperature maps and the experimental results of this internship along with comparing the simulations with the experimental data. The last chapter, chapter 6, concludes the work in this report and discusses the future scope of this project.

Chapter 1

Heat Transport at the Nanometre Scale and Nanothermometry

Heat in a solid is carried by electrons and phonons. Metals have a high density of electrons by which they conduct heat, making them good conductors of heat whereas dielectrics and insulators conduct heat preferentially via phonons, which are the quanta of lattice vibrations. Both carriers are scattered by defects, boundaries and by one another. These heat carrier have a finite mean free path between consecutive scattering events, which is material dependent and affects the thermal transport. This chapter provides an overview of heat transport phenomena and discusses various nanothermometry and microscopy techniques that have been developed to characterize heat transport at the nanoscale. The chapter also highlights the limitations of these techniques and the need for a new approach that can provide high spatial and temporal resolution for studying heat dynamics at a nanometre scale.

1.1 Heat carriers in solids

Heat conduction in a solid is carrier out by mobile carriers present in the material. Two major contributors to thermal conducation are electron and phonons.

- **Electrons** are present in high density in metals and are effectively absent in insulators. They travel at fermi velocity, in the order of $1 \times 10^6 ms^{-1}$ [9]. They are scattered by lattice defects, grain boundaries, phonons and surface of the solid. As electron also carry a charge, the thermal conductivity is often tied to the electrical conductivity.
- **Phonons** are the quanta of lattice vibrations. They travel at the speed of sound, in the order of $1 \times 10^3 ms^{-1}$ [10]. They are scattered by each other, lattice defects, grain boundaries and surfaces.

The thermal conductivity of a solid is the sum of these two contributions, and the total thermal conductivity is given by

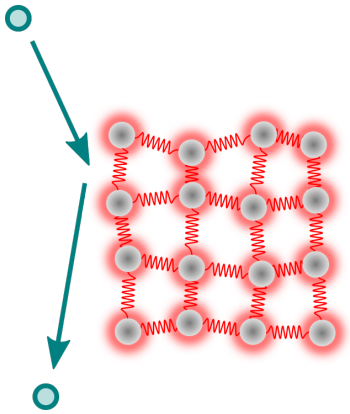
$$\kappa_{\text{tot}} = \kappa_{\text{el}} + \kappa_{\text{l}}, \tag{1.1}$$

where κ_{el} is carried by conduction electrons and κ_{l} by phonons. In a dielectric the first term is negligible as there are almost no free electrons [2] whereas in a metal electronic term dominates, and it is tied to the electrical conductivity σ by the Wiedemann-Franz relation [9],

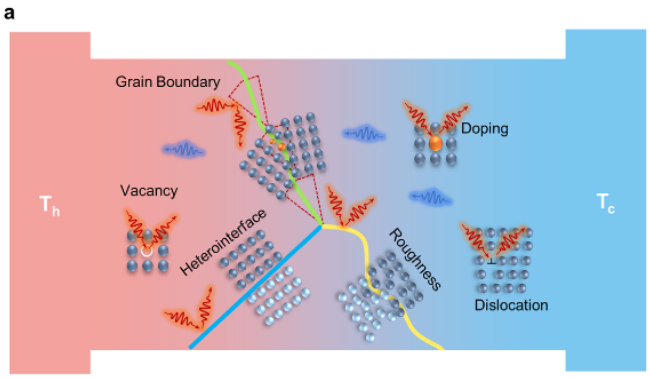
$$\kappa_{\text{el}} = L_0\,\sigma\,T, \tag{1.2}$$

where T is the absolute temperature and $L_0 = \pi^2 k_{\text{B}}^2/3e^2 = 2.44 \times 10^{-8}\,\text{W}\,\Omega\,\text{K}^{-2}$ is the Lorenz number. For aluminium at 300 K, $\sigma \approx 3.66 \times 10^7\,\text{S}\,\text{m}^{-1}$ gives $\kappa_{\text{el}} \approx 268\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$. From first-principles calculations only $6\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$ is attributed to phonons [11]. Thus in a metal the electrons carry most of the heat.

In a crystal lattice the atoms vibrate about their equilibrium positions and a quantum of that vibration is called a phonon. The lattice temperature is described by the total population of these phonon modes. The two thermal carriers interact with each other and their equilibrium is essential to hold the temperature. Figure 1.1 illustrates the various scattering processes.



(A) A scattering process between an electron and lattice vibrations.



(B) Various scattering processes occuring during the thermal transport from a hot to a cold region.

Figure 1.1: Thermal transport carriers undergo various scattering processes during their transport resulting in reduced thermal conductivity. Panel (b) reproduced from Xu *et al.* [6], figure 2(a).

Motion of electrons in a metal is modelled as the motion of free gas particles and the dynamics are derived from the kinetic gas theory. The electrons do not travel in straight lines and undergo many scattering events. The distance a carrier covers between two consecutive scattering events is its mean free path Λ. This transport is known as diffusive transport, where the dimensions of the structure is larger than Λ.

When the dimensions of the structure are shorter than the mean free path, a carrier crosses it without scattering except at the surfaces, and the transport is known as ballistic transport. As the device size shrinks, new scattering processes contribute significantly and affect the overall thermal conductivity of the material. Boundary scattering occurs at the interface and free surfaces, and as a structure shrinks the surface-to-volume ratio increases, increasing the scattering events and reducing the mean free path of the carriers. From kinetic theory we can connect the mean free path to the conductivity. Treating the carriers as gas drifting in a temperature gradient gives [9]

$$\kappa = \frac{1}{3}\, c\, v\, \Lambda, \tag{1.3}$$

where c is the volumetric heat capacity of the carriers, v their characteristic speed and Λ their mean free path. For a metal such as aluminium the transport is dominated by electrons near the Fermi surface, so a single characteristic velocity is the Fermi velocity. Mean free path of a typical metal is in the order of a few nanometres [12]. Spatially, for a structure larger than the mean free path the diffusive thermal transport is described by Fourier's heat law discussed in section 1.3. Temporally, the time scales of transport must be larger than the equilibrium time of the thermal carriers. It is discussed in section 1.2.

1.2 Thermal transport induced by optical excitations

A laser pulse absorbed in a metal does not raise the temperature of the metal in a single step. A part of the energy abosrbed by the electrons is liberated to the lattice and then spreads as heat. The four stages are physically distinct and run from the femtosecond to the hundred-picosecond range.

1. **Electron excitation.** The pulse of light is absorbed by the conduction electrons through band transitions and in a nanosctructure, absorption is enhanced through plasmonic excitations. The electrons are excited far above the fermi level. They do not follow a thermal distribution yet, which sets in after the equilibrium processes.

2. **Electron thermalization.** Electron-electron scattering redistributes the absorbed energy over the whole electron gas. After a few tens of femtoseconds the distribution changes back to Fermi-Dirac and an electron temperature T_e exists, far above the lattice temperature [13]. At the end of this stage the electron system have a temperature.

3. **Electron-phonon interaction.** The hot electron gas loses its energy to the lattice by emitting phonons. The electron-phonon interaction exchanges the heat from the hot highly mobile carriers to slow moving phonons and the time of equilibrium for a metal is spanned in the range of $0.1 - 1\,\mathrm{ps}$ [13].

4. **Phonon relaxation.** The phonons emitted by the electrons are not distributed as a thermal population either: the coupling favours particular modes. Phonon-

phonon coupling exchanges energy over a spectrum over tens of picoseconds and attain a single lattice temperature [14].

Table 1.1: The four stage of optical excitation and thermal carrier relaxarion.

Stage	Process	Time scale	Quantity that exists at the end
(i)	Absorption and electron excitation	pulse duration	a non-thermal electron distribution
(ii)	Electron thermalization (e–e)	10 – 100 fs	an electron temperature T_e
(iii)	Electron–phonon energy transfer	0.1 – 10 ps	a hot, non-thermal phonon population
(iv)	Phonon–phonon relaxation	1 – 100 ps	a single lattice temperature T

All four stages are complete within a hundred picoseconds. A single-temperature Fourier model which works in nanoseconds and beyond is then sufficient.

1.3 Fourier's law and the heat equation

In diffusive thermal transport, heat behaves like a fluid flowing down the temperature gradient. Fourier's law states that the heat flux $\mathbf{q}$, the power crossing unit area, is proportional to the gradient of the temperature,

$$\mathbf{q} = -\kappa \nabla T, \tag{1.4}$$

where κ is the thermal conductivity of equation (1.1), in $\mathrm{W\,m^{-1}\,K^{-1}}$, and the minus sign signifies the heat flow from hot to cold.

The Fourier's law combined with energy conservation is solved to determine the rate of heat flow and the temperature of the element.

$$\rho\, c_p \frac{\partial T}{\partial t} = -\nabla \cdot \mathbf{q} + Q, \tag{1.5}$$

Equation (1.5) is linear in T, so for a simple geometry with constant coefficients it can be solved analytically by seperating the variables and expanding the temperature field in the eigenmodes of the domain, i.e. the Fourier series. Each mode n then decays independently as $\exp(-t/\tau_n)$ with $\tau_n = 1/(Dk_n^2)$, where k_n is the wavenumber of the mode and D is the thermal diffusivity. The long-time behaviour is governed by the slowest mode as the fast modes with large k_n die out first.

A real structure is neither homogeneous nor simple and has several materials, an interface between them, and a source that varies in both space and time. The same equation is then solved numerically by the finite-element method. The domain is divided into a mesh of elements, the temperature field is represented by its values

at the nodes, the spatial derivatives in equation (1.5) are replaced by their discrete approximation, and a large system is constructed by coupled ordinary differential equations forwarded in time steps from the initial condition [15].

1.4 Size effects on heat transport

The conductivity measured for a thin film is not the conductivity of the bulk material it is made of [2, 3] as explained by equation (1.3). The heat capacity c and the carrier velocity v are properties of the material and do not change when a film is made thinner. The mean free path Λ changes as a carrier that would have travelled a mean free path in the bulk now meets a surface first due to the enhanced surface-to-volume ratio, and the conductivity falls in proportion. Other important factors to consider on size reduction are as follows

The Casimir limit. In the extreme case the mean free path is set entirely by the dimensions of the sample. A scattering event between the surface and the thermal carrier is equivalent to scattering events in a bulk as the carrier loses information of any past momentum. This saturates the effectice mean free path at the smallest dimension d of the structure [16],

$$\Lambda_{\text{eff}} \approx d. \tag{1.6}$$

Substituting into equation (1.3) and dividing by the same expression for the bulk gives a conductivity reduced by the ratio of the two lengths,

$$\kappa_{\text{nano}} = \frac{d}{\Lambda_{\text{bulk}}} \kappa_{\text{bulk}}, \tag{1.7}$$

which holds when $d \ll \Lambda_{\text{bulk}}$ [17].

Surfaces and grain boundaries. In a polycrystalline metal film the grain boundaries become a dominent source of scattering events. In a thermally evaporated film, the thickness is limited by the grains. The added grain boundaries decrease the mean free path decreasing the thermal conductivity [18, 19, 4].

1.5 Thermal interface resistance

Heat crossing an interface between two materials encounters a resistance present even when both sides are perfect crystals in direct atomic contact. Carriers arriving at the interface are partly transmitted and partly reflected back due to a mismatch between the vibrational or electronic states in the two materials. This results in a sharp temperature drop localised at the boundary while the flux through it stays continuous, as shown in figure 1.2. Atomically the drop is spread over a few atomic

layers and is treated as a step. The ratio of the drop to the flux defines the thermal boundary resistance, also called the Kapitza resistance,

$$R = \frac{\Delta T}{j}, \tag{1.8}$$

where j is the heat flux across the interface and ΔT the temperature drop across it. R has units of $\mathrm{m^2\,K\,W^{-1}}$, and its reciprocal $G_\mathrm{K} = 1/R$ is the interface conductance. It enters the thermal model as the interface boundary condition of section 1.3, and it adds a resistance in series with the material on either side (see sec. 5.4).

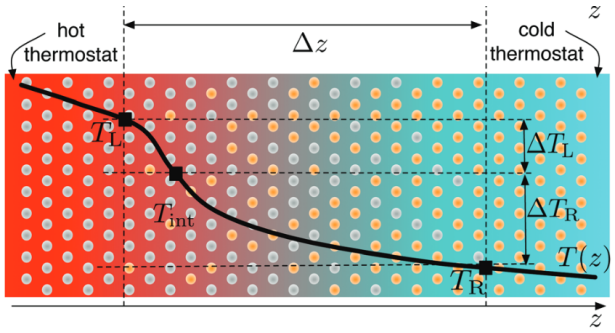
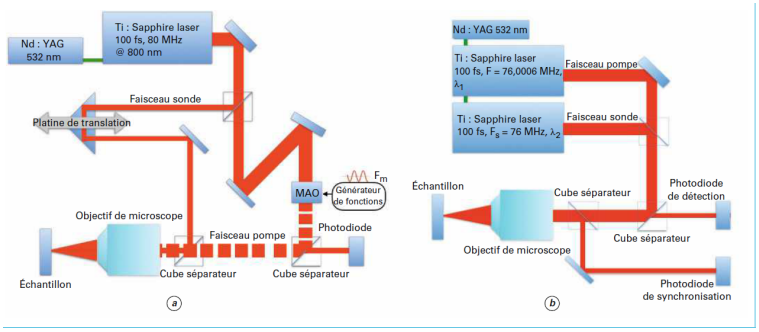


Figure 1.2: Temperature profile across an interface between two solids. The profile is linear within each material and falls steeply across the interface region. Reproduced from Rurali *et al.* [20].

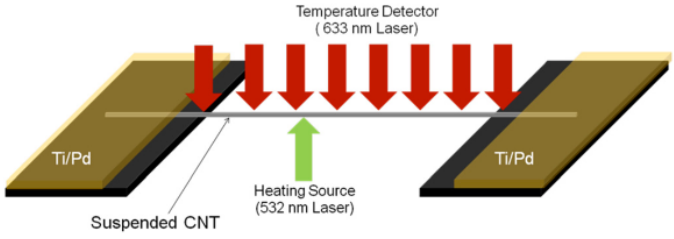
1.6 Nanothermometry techniques

Many techniques measure thermal transport at small scales, each suited to a particular material and geometry. Figure 1.3 shows the configurations of the most widely used among them.

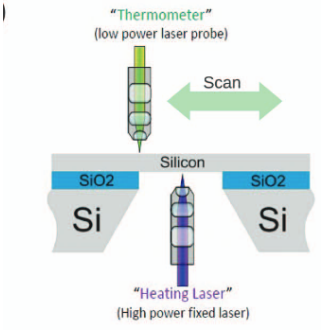
Optical methods. These use the interaction between light and the sample. Raman thermometry (see fig. 1.3b) takes the temperature from the shift of a vibrational mode. The lattice potential is anharmonic, so as the crystal expands on heating the mode softens and its characteristic peak moves to a lower wavenumber; the shift is close to linear over a few hundred kelvin, and a calibration slope measured on a known temperature converts the shift into a temperature. Thermoreflectance instead measures the change in optical reflectivity of the surface with temperature. It is a pump-probe technique: one pulse heats the sample and a second, delayed one reads the reflectivity, and the transient is built up either by scanning the delay (TDTR) or by sweeping the modulation frequency of the pump (FDTR)(see fig. 1.3a) [23, 24]. Thermoreflectance resolves picoseconds, and Raman with a pulsed source resolves nanoseconds. Both are diffraction limited to a few hundred nanometres in space. Thermoreflectance also needs a metal transducer layer, which introduces one extra interface into the measurement. Raman sensitivity is set by the spectral resolution of the spectrometer, typically $0.25\,\mathrm{cm^{-1}}$ [25].



(A) Time-domain thermoreflectance



(B) Raman thermometry



(C) A scanned laser thermometer over a patterned substrate

FIGURE 1.3: Established nanothermometry methods. (a) Thermoreflectance in its single-laser implementation with a mechanical delay line (left) and its two-laser asynchronous implementation (right) [17]. (b) Two-laser Raman thermometry of a suspended nanotube. [21]. (c) measures the specimen's own vibrational mode [22]

Contact and probe methods. Scanning thermal microscopy (SThM) uses an atomic-force tip to map the sample and measure the thermal profile, recovering the spatial resolution [26], but adds an unkown thermal contact resistance. The SThM has a limited temporal resolution of 1 kHz due to the limited bandwidth of the physical cantilever [27].

Plasmon-energy-expansion thermometry (PEET). This method utilises the plasmonic response of a material and measures the variation in plasmon energy to determine the temperature. Mecklenburg *et al.* mapped Joule-heated aluminium wires from the energy of the aluminium bulk plasmon [28]. The bulk plasmon energy is a function of the free-electron density, which varies as the lattice expands on heating (sec. 2.5). The plasmon energy is measured in a transmission electron microscope (TEM) by electron energy-loss spectroscopy (EELS). The reported spatial resolution by Mecklenburg *et al.* is nanometre-scale. The measurement is in steady-state due to lack of the temporal resolution and the map is built up over an acquisition long

compared with any thermal transient, so no dynamics are accessible.

Table 1.2: Nanothermometry techniques and the constraint that limits each one.

Technique	Spatial	Temporal	Principal limitation
Raman	$\sim 0.5\,\mu m$	ns	sensitivity set by spectral resolution
TDTR / FDTR	$\sim 0.5\,\mu m$	ps	diffraction limited; transducer adds an interface
SThM	$\sim 50\,nm$	ms	contact resistance; probe loads the sample
PEET	$\sim 3\,nm$	—	steady state only
This work	$\sim$ 10s of nm	ns	counting statistics / beam current

Figure 1.4 compares a wider set of methods across five axes at once. Most of the optical methods Raman, photoluminescence, fluorescence ecompass a wide set of factors yet lack in spatial resolution. Scanning thermal microscopy and PEET are very good for very particular applications but lack the required temporal resolution. PEET is the extreme case which has the highest thermal and spatial resolution. Modifications to the PEET method can improve the temporal resolution without compromising the spatial resolution. This work aims to extend the PEET method to a new temporal regime, while retaining the spatial resolution (see chapter 2).

1.7 Conclusions

This chapter provided an overview of the thermal transport phenomena in solids (sec. 1.1). We concluded that in the case of metals, most of the heat is transported via electrons and the thermal conductivity depends on the mean free path of these carriers which decreases upon size reduction. We saw the four stages of thermal carrier relaxation which occurs within a hundred picoseconds (sec. 1.2), hence making the Fourier's heat law applicable for devices with dimensions larger than the mean free path of the carriers (sec. 1.3). At the end we concluded that, optical techniques are limited in spatial resolution by the diffraction limit and contact probes are limited by their temporal resolution due to limited bandwidth of the probe (sec. 1.6). To study the dynamic effects of heat transport at a nanometre scale, we need a technique which can resolve both spatially and temporally. Modifications to PEET to enhance the temporal resolution is investigated in this work.

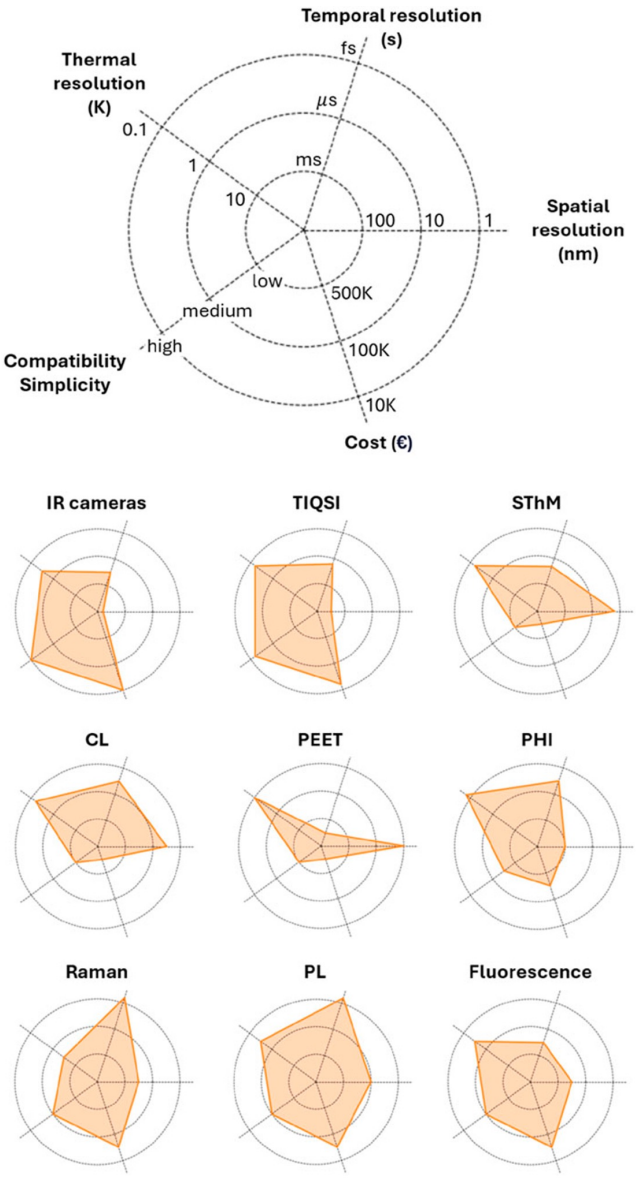


FIGURE 1.4: Nanothermometry methods compared on spatial resolution, temporal resolution, thermal resolution, cost and experimental simplicity. Each axis improves outwards. PEET occupies the right-hand spike: the best spatial resolution of any method shown, and almost no temporal resolution. Reproduced from Romeo *et al.* [29], figure 5.

Chapter 2

Electron Microscopy, EELS and STEM-EELS Nanothermometry

Optical microscopes are widely used in research and industry to evaluate sample quality and infer useful properties of the material. They are limited by the wavelength of light. Abbe's diffraction limit sets the resolution limit of two points [30],

$$d = \frac{\lambda}{2\,n \sin\alpha}, \tag{2.1}$$

with λ the wavelength, n the refractive index of the medium and α the half-angle of the collected cone. Visible light spans 400 - 700 nm limiting the optical resolution to roughly 200nm. To examine finer structures, electron microscopes are used. Fast moving electrons have a much shorter De-Broglie wavelength than visible light [31].

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_0eU\left(1 + \frac{eU}{2m_0c^2}\right)}}, \tag{2.2}$$

with h Planck's constant, p the momentum, U the accelerating voltage and m_0, e and c the electron rest mass, its charge and the speed of light, the last factor being the relativistic correction. An electron microscope working at a typical operating voltage (100keV) has a De-Broglie wavelength it the order of picometres.

An additional advantage of electrons is the interaction of electrons with matter where it exchanges energy with the material which is charaterised by a technique known as the Electron Energy Loss Spectroscopy (EELS). This chapter describes an electron microscope and the modifications made to a standard scanning transmission electron microscope (STEM) to achieve high temporal resolution along with narrow electron energy dispersion. It elaborates the technique of STEM-EELS used in this work and develops the theoretical understanding of relation between the bulk plasmon peak and temperature.

2.1 Electron microscopy

Electron microscopes are divided into two catagories depending on spatial resolution offered by the microscope.

- **The scanning electron microscope (SEM).** It focusses the electron beam onto the surface of a bulk sample, forming a probe with electron energy between 0.5 - 30keV. The electrons scattered from the suface are collected to reconstruct the surface topology. The backscattered electron carry information about the atomic number of the specimen hence they can be used to infer the chemical composition of an unknown sample. As the electrons are scattered from the top surface of the sample, there is no special requirement to thin the sample.

- **The (scanning) transmission electron microscope, (S)TEM.** It focusses the electron beam onto the surface of a thin sample where the electrons are transmitted through the thin sample due to their high energy (fig. 2.1). As the transmitted electron beam has a higher energy typically between 60 - 300keV, it forms a high resolution image of the atomic columns and the internal structure of the sample. In a scanning transmission electron microscope (STEM) the high energy electron beam is scanned on the thin sample forming a high resolution map. The probe formed is sub-nanometre in diameter and moves on the sample point by point. An STEM can form spectral images consisting of one spectrum per scan point

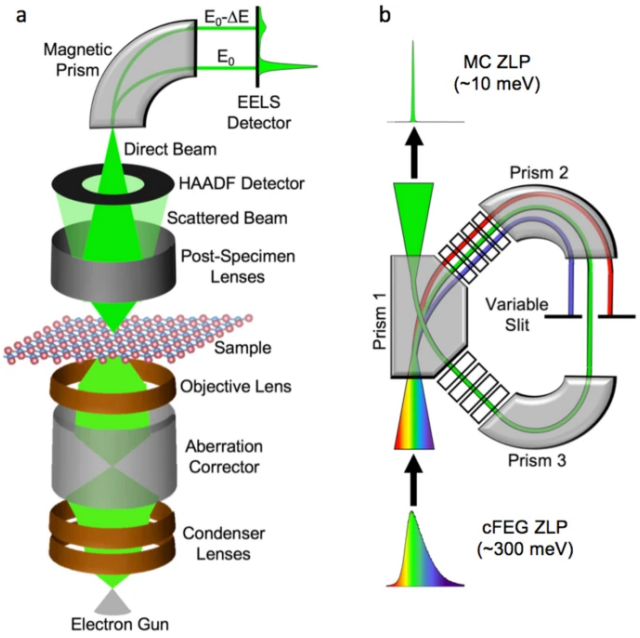


Figure 2.1: The schematic diagram of (a) the electron microscope column and the EELS spectrometer, and (b) the monochromator. Reproduced from Hachtel *et al.* [32]

In a STEM, electron are emitted from the cold field emission tip and accelerated under high voltages (high tension), which produces a bright beam with energy spread of a few hundreds of meV [32]. Electromagnetic lenses are used to focus the beam to form a narrow electron probe. These lenses apply magnetic force on the charged particles to shape the beam. Focussing is especially difficult for the case of electrons as Scherzer showed that a round, static electron lens produce a highly abberated beam [33]. In the case of optics, divergent lenses are used to correct for these abberations but divergent lenses do not exist for electrons. These abberations are corrected by a set of complicated multipolar lens system called abberation correctors. In the most advanced machines, a monochromator (fig. 2.1) can help to reduce the beam energy spread up to $\sim 10meV$, reducing both the chromatic aberrations of the probe and giving access to low excitation energies in spectroscopy.

Scan coils then move the probe across the specimen in a raster scan, and the detectors listed below record the electrons once per position.

Electron beam interaction with the sample and various signals

The electron beam entering the sample undergo many types of scattering processes and each signal provides some information about the type of interaction. These scattered electrons (together with the secondary signals emitted) can then be used to collect information sorted in three categories of acquisition. Imaging detectors use the contrast between the intensity of scattered electrons to form an image. The diffraction modes records their angular distribution on a pixelated camera. Spectroscopy acuisitions measure the energy loss of either the electron beam itself (EELS) or of the secondary signal generated, such as photons (Cathodoluminescence) or X-rays (Energy-Dispersive X-ray spectroscopy) [34]. Table 2.1 provides an overview of these techniques.

For the present work, we use a combination of HAADF imaging and time-resolved EELS spectroscopy acquisition to resolve the temperature evolution in metallic nanostructures.

Table 2.1: The three families of techniques available on a STEM, the signal each one uses and what it measures

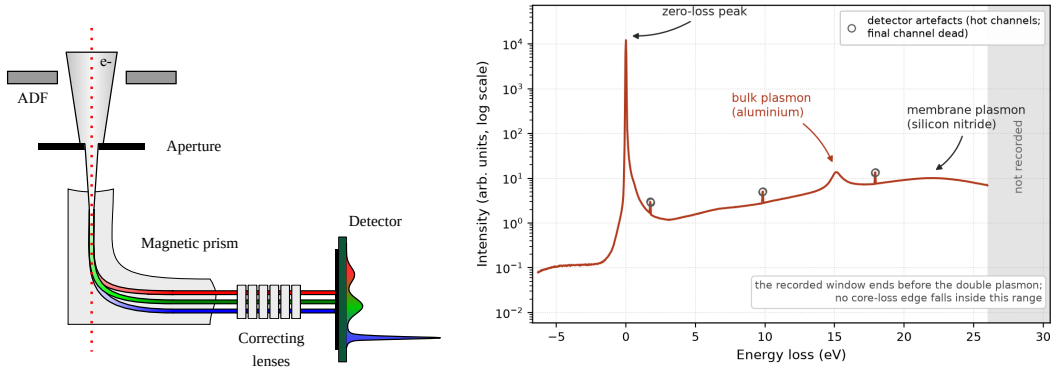
Family	**Signal**	**measurement**
Imaging	direct on axis beam (BF), scattered electrons at higher angles (ADF, HAADF)	lightly scattered electrons detected near the beam axis by BF and heavily scattered electrons are detected by the Dark field detectors.
Diffraction	angular distribution on a pixelated camera	local crystal structure, orientation, strain, electrostatic and magnetic fields in the sample.
Spectroscopy	energy lost by the transmitted electron (EELS); light collected by a parabolic mirror (CL); characteristic X-rays (EDX)	Energies lost by an electron provides information about the type of excitations in the sample; CL gives the photons emitted by the sample under the beam; EDX provides the elemental composition

2.2 Electron energy-loss spectroscopy

A fast moving electron passing through a sample excites various processes and exchanges energy. A magnetic prism after the sample interaction deflects the transmitted beam by the amount of energy it lost (fig. 2.2a). The beam is spread into a spectrum, and its features encode the information about the type of inelastic excitations (fig. 2.2b).

The losses fall into particular regions, each belonging to a different class of excitation.

- **The zero-loss peak (ZLP)**, at zero loss. Electrons crossing the sample without a measurable loss, together with the ones scattered elastically. Its width sets the energy resolution of the experiment.
- **The phonon region**, at a few to a few hundred meV. The beam electrons create or absorb a phonon. These losses or gains are present near the tail of the ZLP and only become accessible with monochromation [35, 36].
- **The valence, or low loss, region**, from about 1 eV to a few tens of eV. The electron beam drives the outer, weakly bound electrons of the solid collectively. It contains the interband transitions and the surface and bulk plasmons.
- **The core loss region**, above about a hundred eV. The beam electrons ejects an electron from the inner shell of the sample atoms, and the onset of each edge is the binding energy of that shell, characteristic to the element [37, 31].



(A) (B)

FIGURE 2.2: (a) Illustration of the EELS prism and spectrometer, showing how the magnetic prism splits the electron beam into a spectrum. (b) A typical EELS spectrum, showing the zero-loss peak, the aluminium bulk plasmon, and the silicon nitride membrane plasmon.

The bulk plasmon

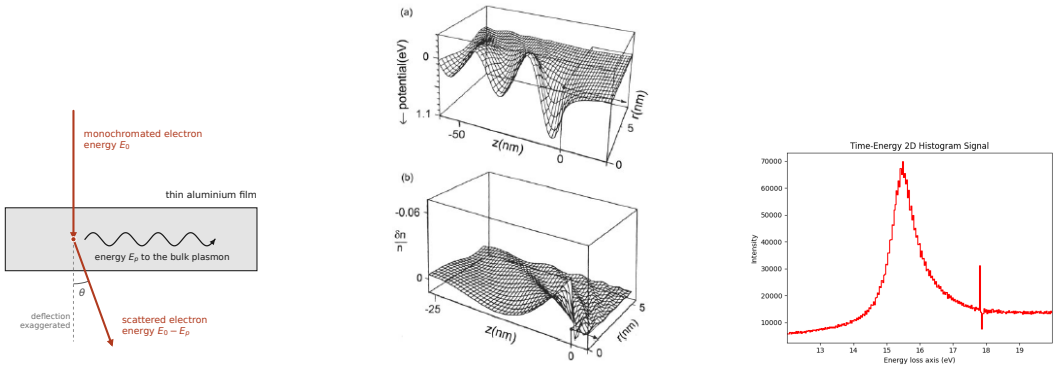
The valence electrons of a metal behaves as a gas of free carriers moving in a rigid lattice. If the charges are displaced, they are pulled back by the counter charges in the lattice sites creating an oscillatory behaviour. It oscillates with a characteristic frequency, the plasma frequency [9]. A bulk plasmon is one quantum of that oscillation. The surface plasmon is bound to a boundary and lies at a lower energy.

A fast moving electron passing through the metal leaves an oscillating wake of charge density behind it (fig. 2.3b). Exciting the bulk plasmon costs the electron a characteristic energy (fig. 2.3a) measured in the EELS spectrum (fig. 2.3c).

In the Drude's model of a free electron gas, the bulk plasmon carries a characteristic energy,

$$E_p = \hbar\omega_p = \hbar\sqrt{\frac{n(T)e^2}{\varepsilon_0 m}}, \tag{2.3}$$

with $n(T)$ the free-electron density, e and m the electron charge and mass, and ε_0 the permittivity of free space. Aluminium has three loosely bound valance electrons per atom and no d band near the fermi level, so it is close to a free electron metal. Its bulk plasmon is strong and narrow (fig. 2.3). A small shift can therefore be measured.

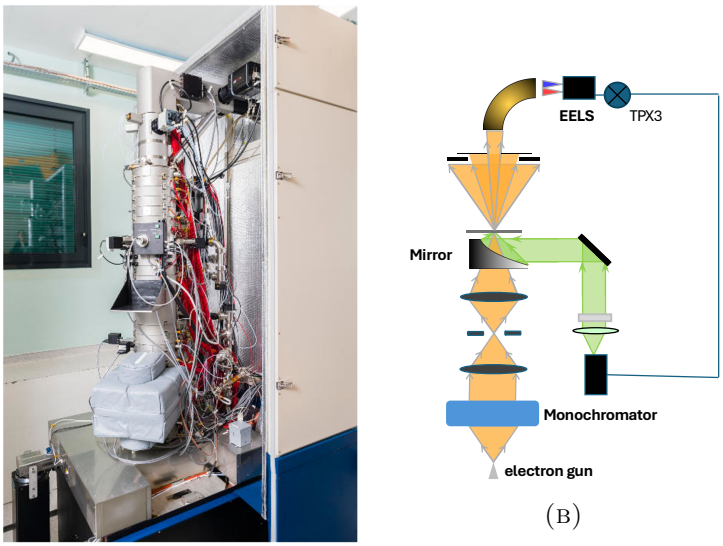


(A) (B) (C)

Figure 2.3: (a) Interaction of the bulk plasmon with incident beam. (b) Wake potential generated in the material. Figure borrowed from Egerton's figure 3.13 [31], after Echenique *et al.* [38].(c) Measured aluminium bulk plasmon of the sample in EELS spectra

2.3 ChromaTEM: a modified STEM

To study the propagation of heat in a nanostructure we need to produce a local hotspot on the sample and measure the evolution dynamically. We also need the ability to resolve small energy changes caused by the temperature change. A standard microscope can not do this.



(A) (B)

Figure 2.4: (a) the ChromaTEM column, and (b) schematic of the ChromaTEM light-injection and detection setup.

ChromaTEM is a modified Nion Hermes 200 at the Laboratoire de Physique des

Solides in Orsay [39, 40]. It is modified with a light injection and focussing mirror setup to produce local hotspots, an event based detector to dynamically resolve the heat evolution phenomena and a monochromator to produce a highly energy filtered beam required to resolve the small energy changes in the EELS signal.

Table 2.2 lists all the modifications along with the requirements of the experiment.

TABLE 2.2: What was added to the base instrument, and the measurement requirement each addition answers.

Modification	**Requirement**	**Feature added**
Monochromator	A cold field-emission gun provides an electron dispersion of 200 - 300 meV which is much larger than the shift to be measured. To enhance the energy resolution of the setup a monochromater is added.	Energy resolution of 30meV was used in this work which was achieved by energy filtering of the original beam. This compromises the beam current as a consequence but provides the required energy resolution to resolve the energy shift due to heating.
Event-based detector	The transient behaviour last a couple of nanoseconds to microseconds. A camera accumulates all the electrons over a longer exposure, discarding the time of arrival of each electron.	Every electron is individually time tagged with a nanosecond resolution giving a time resolved picture of incoming electrons
Light injection port	The sample has to be heated inside the column under ultra high vacuum with high spatial precision.	An optical fibre delivers a pulsed laser injected into the microsope column through a parabolic mirror.

This unique set of modification make ChromaTEM the only setup in the world with the capability of synchronised light injection with beam spot limited by the diffraction limit, time resolved EELS and high energy resolution [39, 40].

2.4 Coincidence spectroscopy and synchronised light injection

To measure the temperature evolution after a pulsed laser heating, which dissipates in a few nanoseconds, we need to collect electron synchronised with the laser pulse. A conventional camera integrates the collected electrons and output a spectrum losing the time of arrival information. The TimePix3 detector developed by CERN [41] registers each electron individually which is corelated to the spatial information of the electron beam with the synchronisation electronics [42, 43, 8]. The ChromaTEM

also collects and injects light pulses which are temporally related to the time of arrival of electrons enabling coincidence spectroscopy.

Light injection

To locally heat the sample, we inject light into the microscope column using a parabolic mirror. The light is produced using a dye laser, guided through an optical fiber, injected into the column and collected by a parabolic mirror (see fig. 2.5a) which is placed on a three axis piezo stage. The stage allows us to change the optical plane and the laser spot location. The laser spot is aligned to the electron probe manually. As the mirror is parabolic, the focal spot is highly astigmatic and deliveres the beam energy in an elliptical shape.

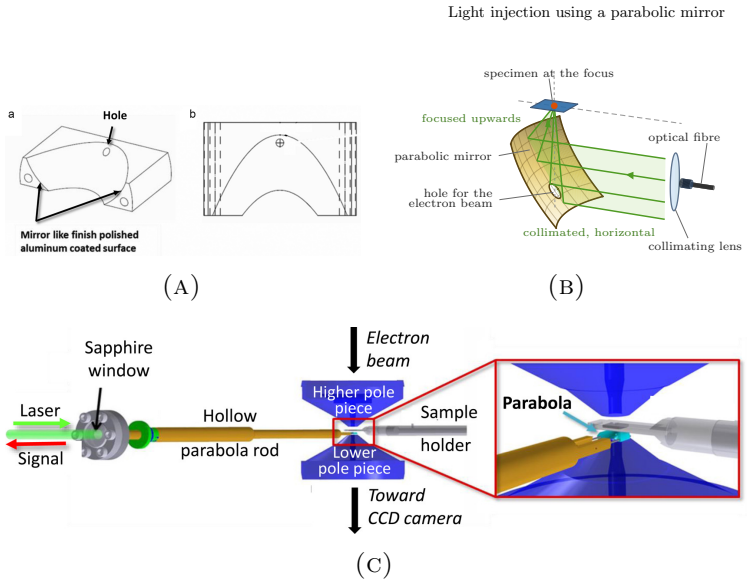
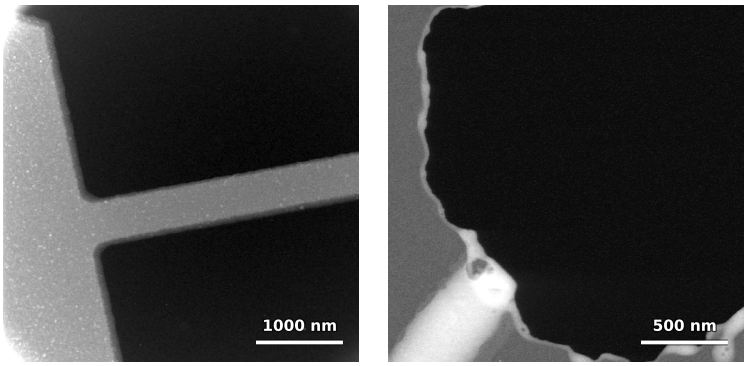


Figure 2.5: Light injection into the electron column. (a) Schematic of the parabolic mirror. (b) Schematic of light injection into the microscope column. (c) Placement of the parabolic mirror inside the electron beam column. Figures (a) and (c) are taken from M. Picher et al. [44].

The injected laser power needs to tuned carefully as the power that produces a measureable temperature rise is close to the power that destroys the structure. This limits the exposure time of the pad as a longer exposure melts and recrystallises the metallic layer modifying the plasmonic response.

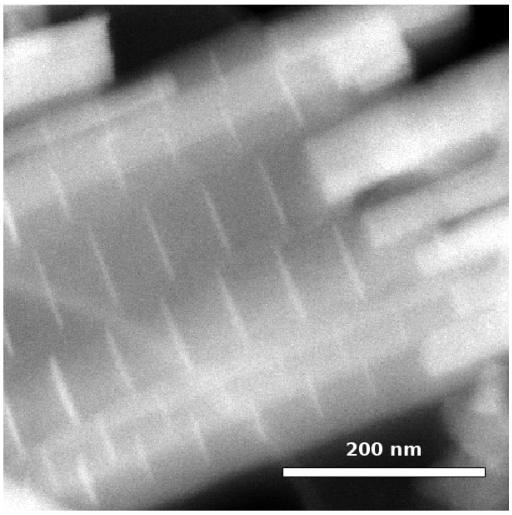


(A) Before illumination. (B) After illumination

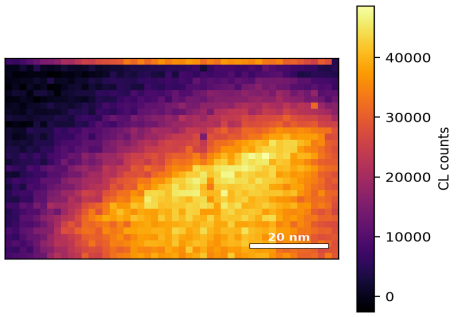
FIGURE 2.6: SiN membrane rupture after excessive power. HAADF images of the same bridge, acquired in the same session before and after laser illumination. The aluminium has balled up at the break, which distinguishes a thermally driven instability from mechanical rupture [45].

Cathodoluminiscence - a tool for mirror alignment

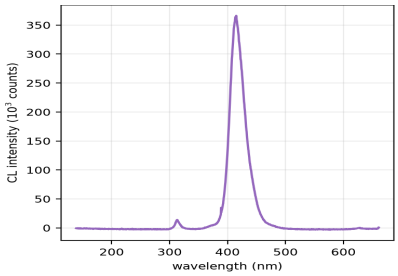
The optical laser is not visible under an electron microscope. The experiment demands a spatial overlap of the electron probe with the laser pump (see fig. 2.8a) within a few hundred nanometres of precesion. The light injection alignment is therefore achieved by the reverse process of cathodoluminiscence. The electron beam excites the light emission in a cathodoluminiscent sample (such as semiconductor nanowires, NV centres) and the mirror is moved to record maximum intensity of CL. As the light is being injected by the same mirror as it is being collected, the laser beam follows the same optical path. Figure 2.7 shows CL alignment on semiconductor nanowires performed on another STEM - the VG Cold present in the group. Figure 2.7b shows a hyperspectral map of an individual quantum well. For this work, we used NV defect centres in nanodiamonds as a CL source to align the mirrors of the ChromaTEM.



(a) ADF image of the nanowire cluster



(b) A hyperspectral image of a single quantum well



(c) Summed intensity of a single quantum well

Figure 2.7: (a) ADF image of a GaN - AlN nanowire cluster, (b) hyperspectral CL image of a single quantum well, and (c) CL intensity spectrum of a single quantum well, used to learn the beam-laser alignment technique.

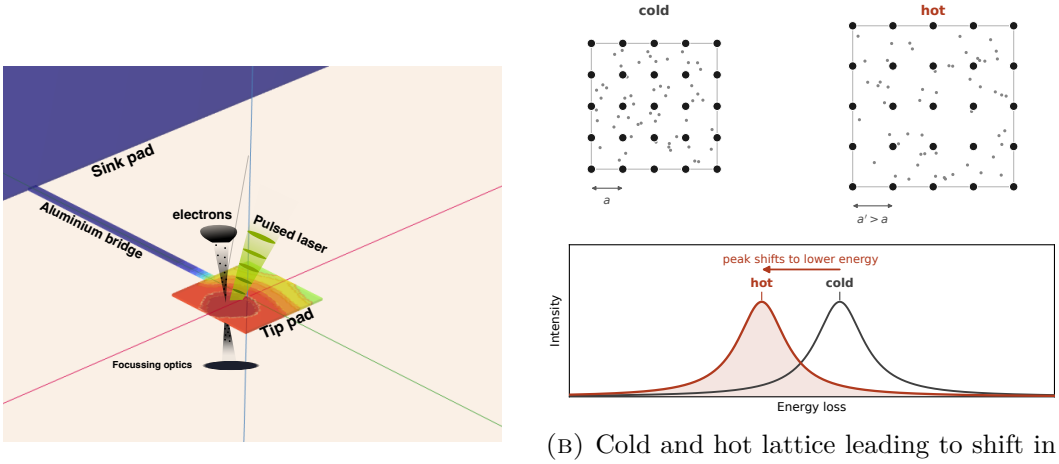
2.5 STEM-EELS nanothermometry

The purpose of this study was to measure the temperature of a nanostructure after a laser pulse. The microscope only records a spectrum, so the temperature has to be extracted from a change in the spectrum. This technique relies on measuring the shift in characteristic excitation energies inside a material. It is not limited to the bulk plasmon.

- **Detailed balance.** At low excitation energies, a fast electron beam adds a fraction to either gain or loss energies when interacting with a material. The ratio between these two signals is related to the absolute temperature of the material, which can then be measured without need of calibration. [46, 47, 7].
- **Band gap modification.** In a semiconductor the gap narrows as the crystal expands, and the onset of the interband loss moves with it [48].
- **Plasmon energy.** In a metal the bulk plasmon energy follows the free electron density [28].

A general overview of the technique can be found in the work by F.Castioni *et al.* [7]. This work uses the third route of plasmon energy.

Heating the sample expands the lattice which decreases the free electron density in the region, lowering the plasmons energy. Figure 2.8 depicts this phenomenon.



(A)

(B) Cold and hot lattice leading to shift in peak.

FIGURE 2.8: (a) Schematic of the STEM-EELS setup alignment. (b) Cold and hot lattice leading to a shift in peak position.

Since the number of conduction electrons is fixed, the density is dependent on the volume (see suppl. [7]),

$$n(T) = n(T_0)\left[1 - 3\int_{T_0}^{T} \alpha(T')\,\mathrm{d}T'\right], \tag{2.4}$$

with α the linear coefficient of thermal expansion. Combining this with equation (2.3) and expanding the square root gives,

$$\frac{\Delta E_p}{E_p(T_0)} = \frac{3}{2}\int_{T_0}^{T} \alpha(T')\,\mathrm{d}T', \qquad \Delta E_p \equiv E_p(T_0) - E_p(T). \tag{2.5}$$

For aluminium α is represented in the quadratic form in $\Delta T = T - T_0$,

$$\alpha(T) = \alpha_0 + \alpha_1 \Delta T + \alpha_2 \Delta T^2, \tag{2.6}$$

with $\alpha_0 = (2.299 \pm 0.016) \times 10^{-5}\,\mathrm{K}^{-1}$, $\alpha_1 = (1.63 \pm 0.13) \times 10^{-8}\,\mathrm{K}^{-2}$ and $\alpha_2 = (8.2 \pm 2.3) \times 10^{-12}\,\mathrm{K}^{-3}$, referred to $T_0 = 27\,^{\circ}\mathrm{C}$ obtained by Chmielewski *et al.* [49, 50].

Carrying out the integral for heating ($T > T_0$) gives,

$$\frac{2}{3}\frac{\Delta E_p}{E_p(T_0)} = \alpha_0 \Delta T + \frac{\alpha_1}{2}\Delta T^2 + \frac{\alpha_2}{3}\Delta T^3. \tag{2.7}$$

Chapter 4 inverts equation (2.7) for each pixel to derive the temperature maps. Near room temperature its leading term dominates, and the sensitivity is

$$\frac{\mathrm{d}E_p}{\mathrm{d}T} \approx -\frac{3}{2}\alpha_0 E_p \approx -0.52\,\mathrm{meV\,K}^{-1}. \tag{2.8}$$

A temperature rise of 100 K therefore moves the peak by about 52 meV, further emphasising the need of a monochromater (see sec. 2.3, tab. 2.2).

2.6 Conclusions

This chapter provided an overview on electron microscopes and various detectors used for microscopy and spectroscopy (sec. 2.1). It elaborated specifically on the electron energy loss spectroscopy (EELS) (sec. 2.2), used to study the shift in bulk plasmon peak. To study the dynamic thermal behaviour, modifications to a standard STEM (the ChromaTEM) (sec. 2.3) were discussed along with the requirements of the experiment. This chapter explained the principle of coincidence spectroscopy and synchronised light injection (sec. 2.4), elaborating on techniques used in the ChromaTEM to achieve this. Lastly, this chapter developed the theoretical framework of the STEM-EELS nanothermometry technique (sec. 2.5) and explained the relation betweem the bulk plasmon energy shift and the corresponding temperature change.

Chapter 3

Simulations, design and fabrication

The primary objective of this work was to investigate the heat transport in a nanostructure. This is the prolongation on the previous work by the STEM group [7] where they performed STEM-EELS nanothermometry on an aluminium thin film. This work extends the technique to nanostructures and presents a possibility to observe heat transport. For that, the geometry of a simple nanobridge, with constricted lateral size to force the heat to propagate through the confined space along a given direction has been opted. The technique being new to the group, no designs or specimens existed before. A simple nanostructure design of a metallic material to ensure a good spectroscopic analysis using the plasmon signal was designed. This chapter catagorically records the development process of the aluminium nanobridge designs and the finite element simulations preformed in COMSOL to test the viability of the designs. The lithography process was especially challenging as it was performed on a 15nm SiN membrane requiring further optimization of the lift off process.

Simulations replace the trial and error of multiple designs and as fabrication is expensive, decreases the cost of development. The key requirement of the design is to maximize the heat flow through the nanobridge from the heating pad towards the cold sink pad. The design was optimized for this reason. These COMSOL simulations also evaluates the effect of the SiN membrane as a potential heat dissipation channel. Simulations also provide a framework to test the viability of the recorded experimental work. A broader evaluation can be seen in chapter 5.

3.1 The finite element method

The simulations were performed in COMSOL Multiphysics 6.4, using the *Heat Transfer in Solids* interface of the Heat Transfer Module, in three dimensions and in the time domain.

Finite element simulations are ideal to solve partial differential equation like the Fourier's heat equation which cannot be solved analytically for certain geometries. The complicated structure is divided into a mesh of tetrahedra with varying mesh

size (tab. A.3). The partial differential equations are solved for each individual element, in this case the temperature evolution is solved in time. The complicated structure is rebuild by the finite elements to calculate the overall behaviour of the structure. A smaller step size constitutes to higher number of elements making the simulation more accurate but increasing the simulation cost. Material properties for the simulation were taken from the COMSOL materials library [51] (tab. A.4).

The COMSOL model solves the classical Fourier's conduction equation (see eq. 1.5). Although it does not model the nanoscale effects discussed in section 1.4, but provides a good starting point for the study.

The laser is simulated as a pulsed gaussian heat source confined to the smaller tip pad area out of which a fraction is absorbed. The intitial temperature of the whole system is set to ambient room temperature. The radient losses are assumed to be negligible as the temperature rise is relatively low for any significant radiation loss. The convection losses are zero in the ultra high vaccuum envirnment of the microscope. Heat is transported only via conduction. The details of the simulations and the parameters are listed in appendix A.

3.2 Design modifications and evolution

The Fourier's heat equation for a uniform thin film can be solved analytically (see sec. 1.3). The solution is an exponential decaying with distance. For a thin film of edge length $20\mu m$ the time scale associated with heat dynamics in the nanosecond range. The nanosecond resolution of the timepix3 detector can be used to investigate this phenomena closely (see sec. 2.4).

The heat in a uniform thin film dissipates in all directions (fig. 3.2a) and the thermal cross section grows symmetriaclly outwards. To enforce a preferential heat flow direction we introduce a thermal bridge between the two pads (fig. 3.2b). The large $20\mu m$ pad dilutes the heat propagating to the nanobridge so size of the pad is decreased to $1\mu m$.

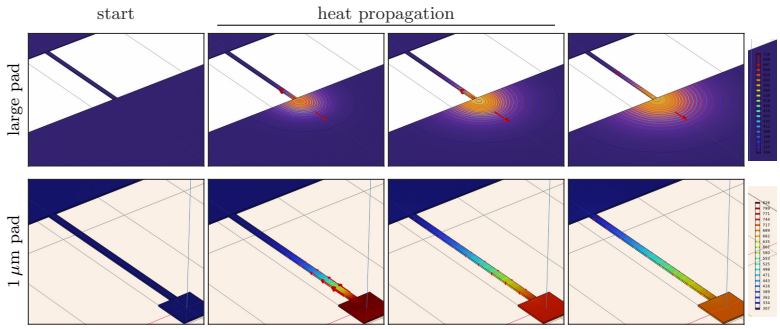
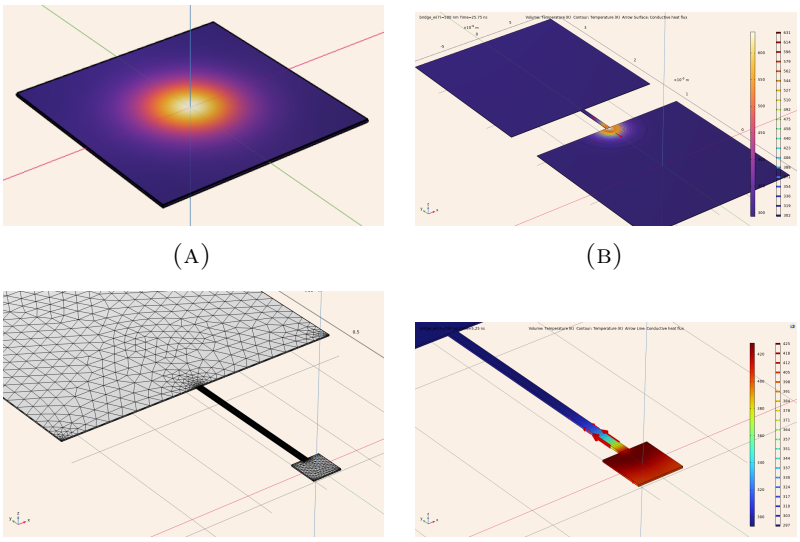


Figure 3.1: upper: heat dilution to the target pad under laser illumination. lower: Uniform heating of the smaller tip pad and directional heat flow to the nanobridge.

Figure 3.1 shows freeze frames from the simulation of the large square pads and the small tip pad. The smaller tip pad heats up uniformally, dissipating the heat

towards the bridge.



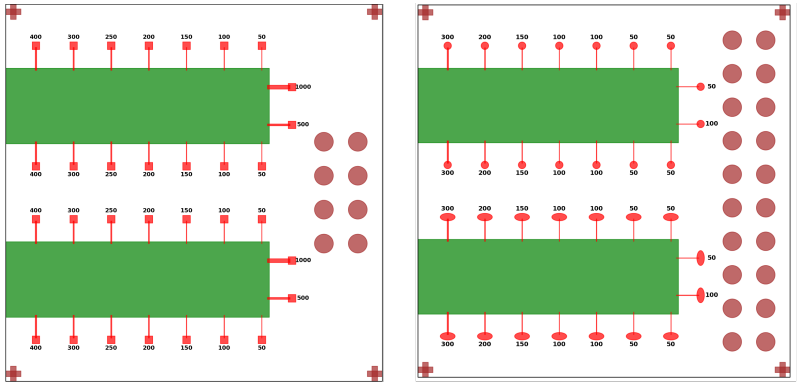
(A) (B)

(C) (D)

FIGURE 3.2: The design evolution optimized for heat propagation through the nanobridge. (a) Uniform thin film shows symmetric dissipation. (b) Equal square pads with laser illumination near the bridge. (c,d) Small tip pad design: (c) finite element mesh with smaller elements in the bridge and coarser in the sink pad (see tab. A.3), (d) heat map of the smaller tip pad design.

3.3 The final design

Final pattern were designed from a practical and experimental point of view produced for repeatability and redundancy. Figure 3.3a shows the first generation design of one sink pad in green connected to the Si frame. A series of nanobridges with varying width are attached to the central pad and replicated on the top and the bottom edge of the sink pad for redundancy. The circular calibration pads to the right are added to align the laser with the electron beam by destructive heating of the pads. They are illuminated with excessive power to create a hole which is used to align the mirror with the electron beam. These pads are used to calibrate the laser power as well (see tab. D.1, appendix D) such that we do not melt the thin metal layer (see fig. 2.6). Figure 3.3b shows the second generation design used in the final experiment. The square tip pads are adjusted to match the astigmatic shape of the injected laser pulse (sec. 2.4).



(a) (b)

Figure 3.3: Final design used (a) first generation square tip pad designs, (b) second generation elliptical tip pad designs

3.4 Fabrication

To realise these structures on a TEM compatible support, the e-beam lithography (EBL) was performed on a standard 15nm thick SiN grid frame (see fig. 3.4a). Reducing the support thickness leads to a fragile structure, prone to rupture during the various fabrication steps. The thin structure also affect the way the electron beam interacts with the e-beam resist, as discussed in section 3.4.1. The methodology for patterning metallic structures on these thin windows had to be established from scratch during this internship.

The EBL was performed using the Zeiss Supra 55 VP SEM machine. Figure 3.4a shows the TED PELLA product no. 21569-10, 3.05 mm TEM lamilla with Si frame and a 3×3 array of 100 µm SiN windows seperated 350 µm apart. Each window hosts a (15±3) nm SiN membrame. Every window is inspected under an optical microscope before processing to ensure the cleanliness and the presence of 9 membranes.

Process flow

Schematics of the process flow is explain in figure 3.4b.

1. **Substrate** – The 3 mm grid is mounted on a silicon wafer with a minimal quantity of silver paste, one edge of the array is kept parallel to the wafer edge so that write-field alignment in the EBL process starts close to correct.

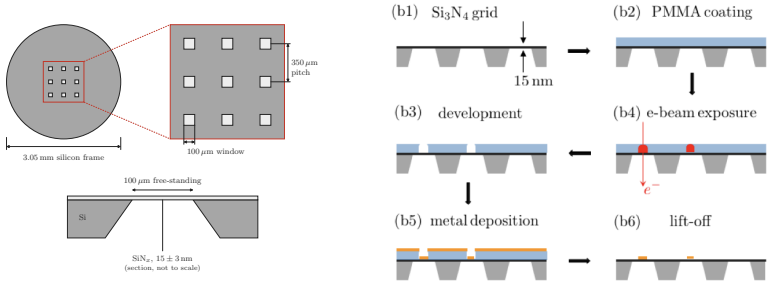
2. **Resist** – Prebake the carrier wafer and the pasted lamila at 180 °C for 15 min. PMMA 950 A3, spun at 3000 rpm for 60 s with 2000 rpm s^{-1} acceleration, giving roughly 150 nm thick film[52]. Post-bake 180 °C for 4.5 min. The more concentrated A6 formulation was tested first and cracked the membrane during post bake process; the 3 % solution gave a uniform resist film.

3. **Exposure** – At a high tension of 20 kV and the 10 µm aperture to limit the beam current to 62.3 pA. Area dose of 600 – 650 µC cm^{-2} was used in the final design.

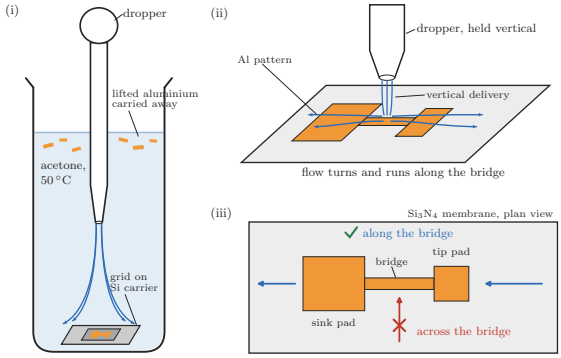
4. **Development** – Washed in MIBK:IPA (1:3) with continuous agitation for 60 s with a rinse in pure IPA to immediately hault the development process.

5. **Metallisation** – Thermal evaporation of aluminium with film thickness of 24.8 nm (targetted at 50nm, cut short due to technical failure). At a pressure of 1.15×10^{-6} mbar, a current of 190A and a deposition rate of $3 \mathring{A} s^{-1}$

6. **Lift-off** – Perfomed in acetone at 50 °C for 10 min with a special two beaker process borrowed from the lab. Refer to figure 3.4c for more detail (see appendix B, tab. B.1).



(A) (B)

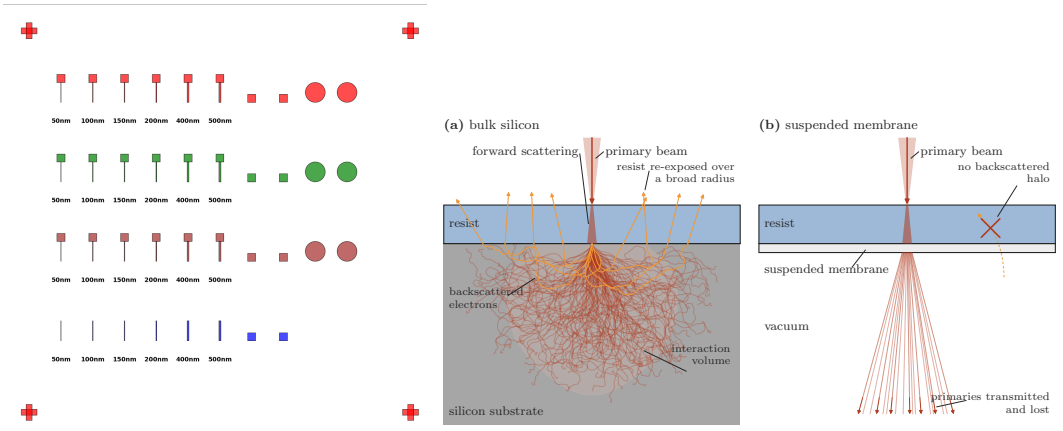


(C)

FIGURE 3.4: Fabrication steps and the special lift off method. (a) The TEM compatible membrane used - drawn to scale. (b) Fabrication flow steps, in order. (c) The special lift off process. Acetone is gently pushed vertically on top of the grid to loosen the deposited aluminium. A laminar flow is maintained to avoid any stress on the thin bridges. The larger flakes are fished out using tweezers. The system is not agitated ultrasonically to protect the delicate SiN membrane.

3.4.1 Dose calibration and test

In a solid substrate the incoming electron are backscattered creating a wide electron cloud (see fig. 3.5b) behind which reacts with the PMMA polymer chains to break it into small monomers which are carried away during the development process. EBL on a suspended membrane behaves differently from lithography on a solid Si wafer. The membrane is nearly transparent at 20 kV where most of the primary electrons pass straight through it and only a few backscattered electrons are produced. This decreases the presence of electron cloud responsible to cause the PMMA polymer breakdown producing sharp isolated features but requires a higher dose than normal. It reduces the proximity effect [53] making it easier to fabricate thin structures but requires an optimization of the beam dose.



(A) (B)

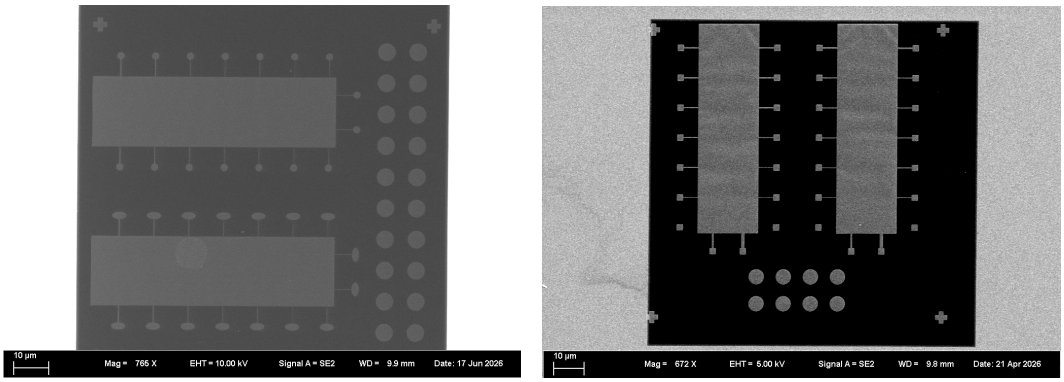
Figure 3.5: Dose calibration test on SiN membrane. (a) dose calibration design used with varying dose in each row. (b) Backscattered electron cloud simulations of a SEM on a solid substrate and a thin SiN membrane.

The design in figure 3.3b is also patterned at varying beam dose of 600 μCcm^{-2} in the top row and 650 μCcm^{-2} in middle and bottom. Within a single design, different layers use different fractions of the nominal dose. The larger sink pads are fabricated at a smaller fraction of the nominal dose than all the smaller features as a larger pattern often benefits from the proximity effect making it ideal for a low dose exposure. After several exposure rounds and tests the relative dose of these sublayers was tuned (see fig. B.1).

3.4.2 Fabrication output

The fabrication process was carried out for both the designs discusses in section 3.3. The patterend aluminium was inspected under the SEM.

An offset in the design and mismatch in the scale of the pattern can be observed in figure 3.6 under an SEM. Figure 3.7 measures the bridges and tip pads at working magnification. The bridge widths were measured from the HAADF images of



(A) Elliptical tip pad geometry (B) Square tip pad geometry

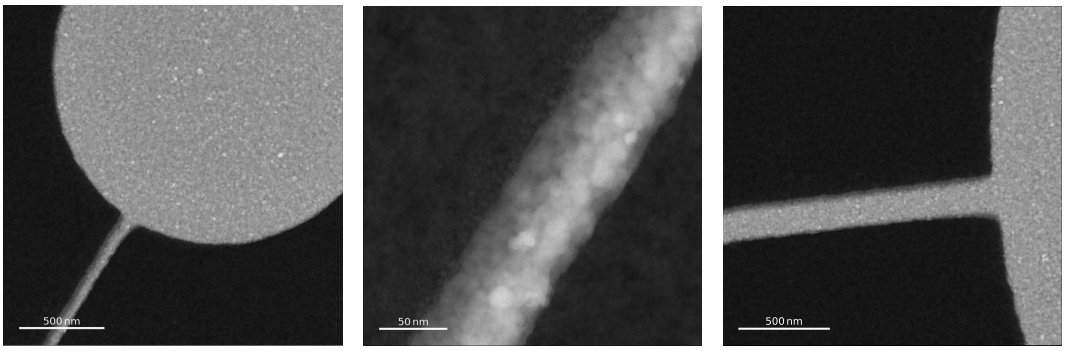
FIGURE 3.6: (a) refers to the second generation design discussed in figure 3.3b. All bridges are cleanly fabricated. (b) refers to the first generation design discussed in figure 3.3a. The 50nm bridges are not patterned in this process.

figure 3.7 rather the SEM inspection images. A detailed analysis is available in appendix C.2. Table 3.1 records the measured widths of the nanobridges. The spread is the scatter among cross section profiles taken at different positions along the same bridge. The variation along the bridge refers to the increase and decrease along the bridge with the measured width which is not the same as fitting uncertainty.

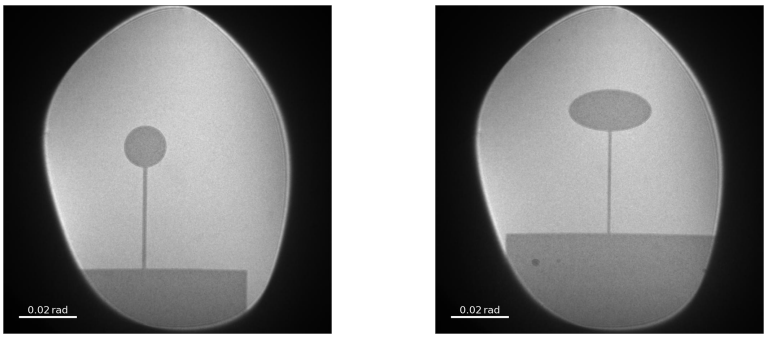
TABLE 3.1: Measured width of fabricated nanobridges with the variation long the brige length

Design	measured width	Variation along the bridge
50 nm	(59.9 ± 1.6) nm	5.2 nm
100 nm	(118.4 ± 2.0) nm	6.6 nm
150 nm	(174.4 ± 2.1) nm	6.9 nm

All the bridges are wider than the target width. Figure 3.7b and the analysis in appendix C.2 show that the bridge is made of a chain of grains whose size differ in width. Evaporated aluminium on nitride grows with grains of a few tens of nanometres, so the narrowest bridges are only a grain or two across and their edges are set by the nucleation spots of individual grains rather than the e-beam resist edge. This is why, the thinnest bridge does not have a flat top and no clean edges. The numbers in the table is a mean over the span covered in the HAADF image. The widest 150nm bridge on the other hand is the cleanest with sharp and well resolved edges with a flat top. Rather than a issue in the EBL process, the surface roughness is an issue of the metallisation process.



(a) (b) (c)



(d) (e)

Figure 3.7: (a)-(c) are STEM-HAADF images. The HAADF signal rises with mass-thickness, so the aluminium is bright and the membrane appears dark. (a) A bridge of the 50 nm design meeting the circular edge of its tip pad. (b) The same design at high magnification showing the evaporated granular film and the bridge edges follow the grains. (c) A bridge of the 150 nm design running into the sink pad. (d) and (e) are bright field images taken over a larger field of view. The aluminium scatters intensity out of the transmitted disc and is therefore darker than the membrane around it, the reverse of (a)-(c), and the bright boundary is the edge of the illuminated area of the difraction plane. (d) shows a circular and (e) an elliptical tip pad, the two shapes designed in the second-generation layout of figure 3.3b.

3.5 Conclusions

This chapter provides an overview on the COMSOL simulations performed to study the heat transport dynamics across a nanobridge (sec. 3.2) and design a pattern which can directionally transport heat through the nanobridge. The fabrication process flow used in the e-beam lithography of nanobridges on a TEM compatible membrane was established (sec. 3.4) and the dose optimisation was performed to find the correct dose for the EBL process (sec. 3.4.1). The fabrication showed clean nanobridges, on the SiN membrane (sec. 3.4.2). The width of the nanobridges is measured to be higher than intended, yet sufficient to conduct the study.

Chapter 4

Data acquisition, structure and processing

The ChromaTEM only records incoming electron hitting the EELS detector. Each electron that hits the TimePix3 camera is tagged with additional labels like, detector column it landed on, the clock tick at which it arrived and the scan position the probe was held when the electron left the specimen (see sec. 2.4). This creates a unique identification for each electron landing on the detector. Each acquisition produces about 300 - 400 GB of data.

This chapter explains the data processing and analysis pipeline in detail to convert the raw data into temperature maps (see fig. 4.9), section 4.1 provides the flowchart of the data processing pipeline, and section 4.3 elaborates each step in detail. Section 4.2 explain the data structure and the in-house `pytpx` library.

4.1 The data processing pipeline

The complete data analysis pipeline developed in this work is shown in figure 4.1. This pipeline can be applied to any set of nanothermometry data acquired to generate the temperature maps from raw .tpx3 data files. Stages 1-5 preprocessses and calibrates the axes, identifying the relevant parameters for the subsequent steps (PINEM: see sec. 4.3.2). Stage 3 is only required, when the spectrometer moves during the scans. Stage 6 crops and rebins the data to build the counting statistics. Stages 7-11 filters and denoises the data to extract the temperature maps for each spatial pixel (see fig. 4.9).

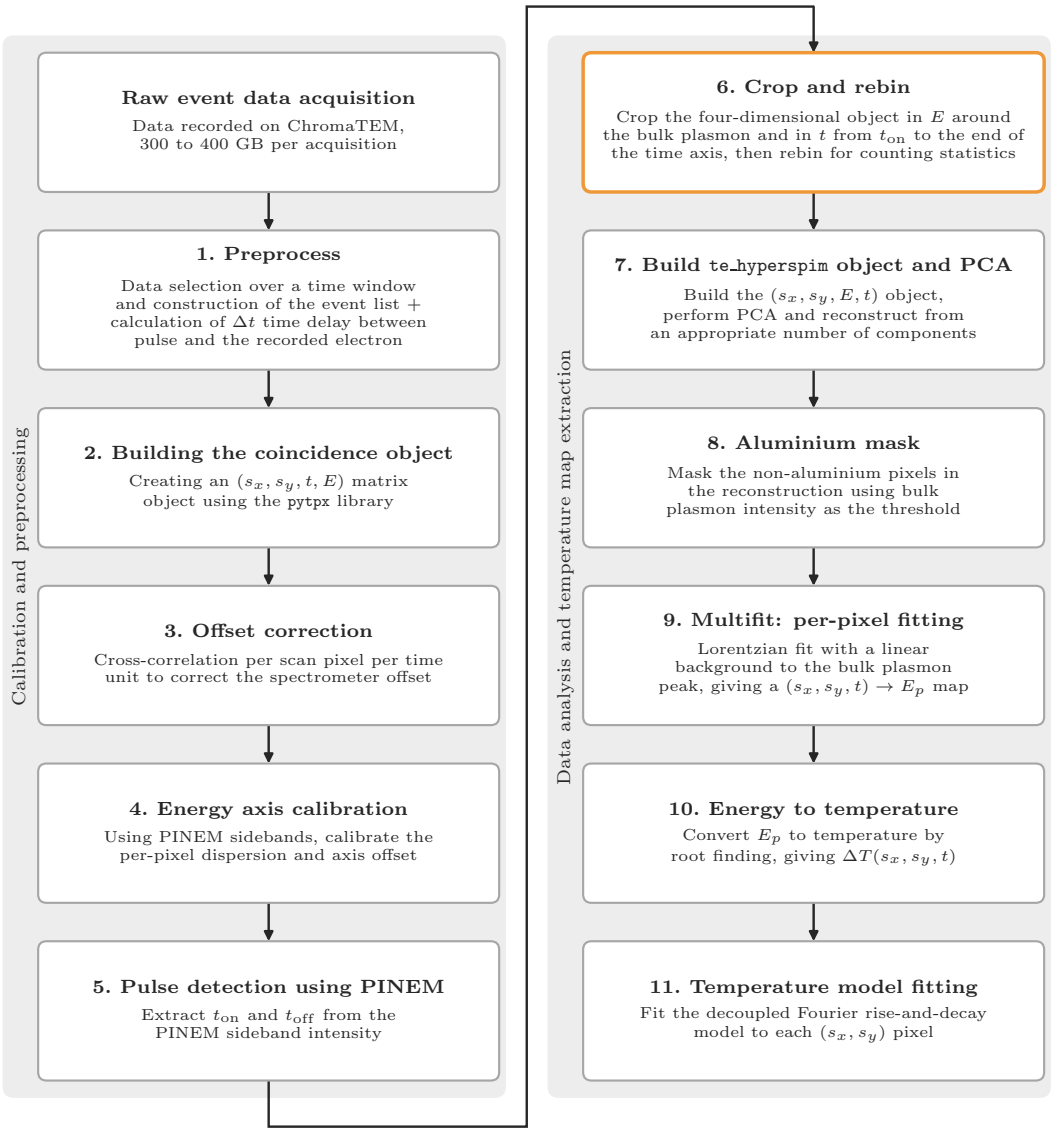


Figure 4.1: Flowchart of the data analysis pipeline

4.2 Data structure and acquisition

In a conventional raster scan, the probe moves across the sample with a spatial step covering the entire scan area. The incoming electrons form a energy loss spectrum and the data is stored a three dimensional array with (s_x, s_y, E) coordinates. In an event based scan the time of delay between the laser pulse and the electron arrival is stored of each electron with these three coordinates[39]. The recorded set of points in (s_x, s_y, E, t) with t referred to the time delay. This enables ChromaTEM to store electrons time tagged to the time delay with respect to the laser pulse.

This information is stored in an ordered event list which stores entries per detected

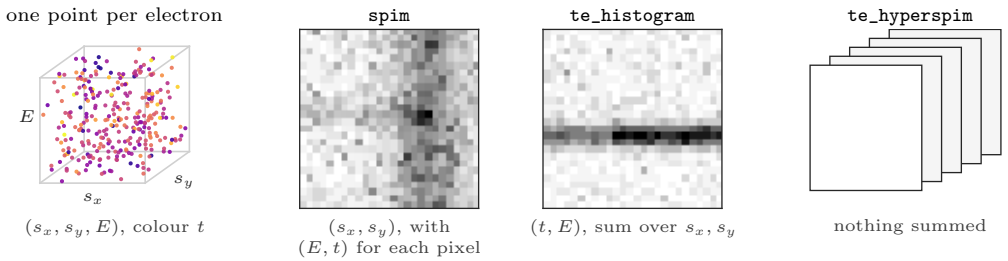


FIGURE 4.2: The event-mode data cube. Each point in the (s_x, s_y, E, t) reflect a stored electron, colour encodes energy loss. The other three panels to the right are dense projections built from that list using the Pytpx library. The scan image summed over energy and time, the time-energy histogram summed over the scan, and the full four-dimensional cube called `te_hyperspim`.

electron rather than per position. The electron are binned together over the repeated laser pulses, building a stroboscopic picture of the experiment. The size of the array is given by the number of electron arriving at the detector, rather than the size of space they live in.

The `pytpx` objects

The in-house `pytpx` library developed by Dr. Florian Castioni is centeral to this work for data processing and coincidence object creation. `pytpx` transforms the sparse data (one event per electron) into a densed matrix format (histogram of events with chosen bin size) enabling an easier workflow. The `spim` for the scan image, `delay_spectrum` and `coincidence_spectrum` for the arrival-time and energy-loss histograms, and `te_histogram` for the time-energy histogram in figure 4.2 are all constructed in this framework.

`te_hyperspim` object is central to this analysis as it contain all the coordinates required to map each electron to its subsequant spatial, temporal and energy pixel. It is the full four-dimensional histogram, (s_x, s_y, E, t), delivered as a HyperSpy `Signal2D` with the two scan coordinates as navigation axes and (t, E) as the signal. Hyperspy[54] is another python library used for data visualisation and data processing. Each spatial pixel has its own time-energy histogram associated which is used to derive the shift in energy from, and later convert to temperature vs time graphs. Building the `te_hyperspim` with its native size is impossible as it is computationally too intense requiring appropriate rebinning and cropping [43, 39].

Acquisition limits

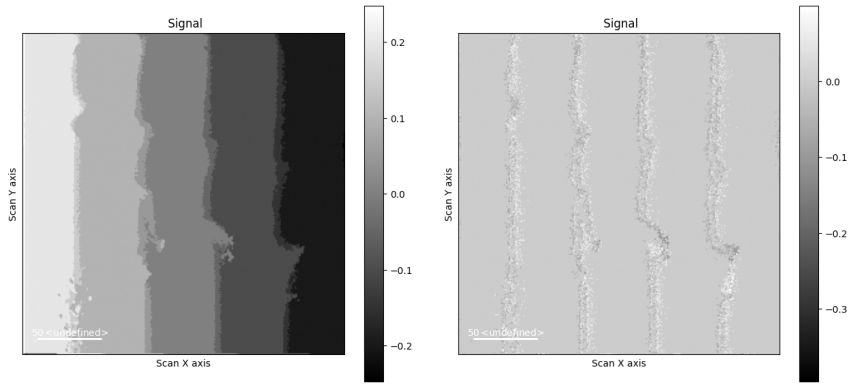
Counting statistics pose the biggest limit on the data acquired. The datasets were acquired with 200 cycles of beam scans over a region of 256×256 pixels with 128 μs dwell time (see tab. D.1, appendix D). The signal was too noisy to fit any function,

pressing the need of rebinning. Data acquisition over more cycles is challenging as the sample drifts slightly over time creating a convoluted signal.

4.3 The data processing in detail

4.3.1 The spectrometer offset correction

Sometimes due to technical errors in the data acquisition process the EELS spectrum on the right side of the frame is shifted in energy with respect to the left side of the frame, creating an offset between the right and the left side of the frame. This is seen as the staircase offset pattern in figure 4.3.



(a) Before correction. (b) After correction.

Figure 4.3: Offset in the EELS spectrum shown as a frame map. (a) Staircase like offset map of the entire acquired frame. (b) Offset map after offset correction. Residual present at the boundries.

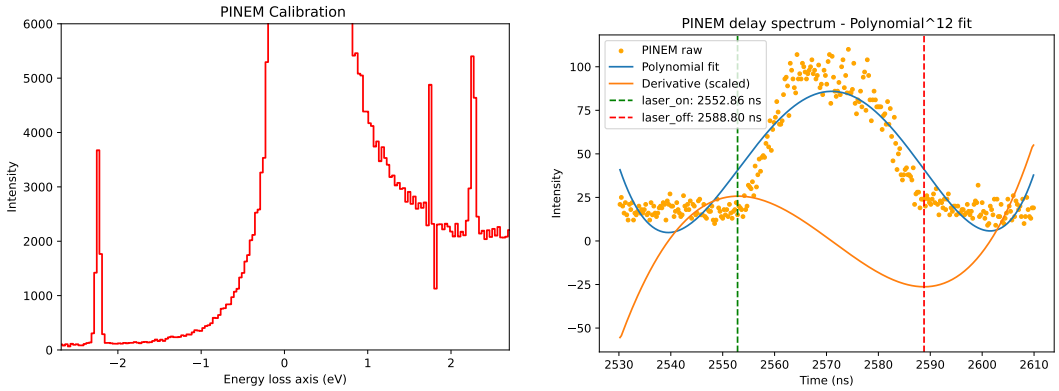
If the offset is present, we extract the shift in bulk plasmon for each pixel which is used to find the relative change in the temperature for that pixel. Presence of an offset produces incorrect temperature maps, needing a correction. The offset maps are generated by cross correlating the EELS spectrum for each pixel with a reference spectrum. The reference spectrum is taken from a small central region of the scan itself binned to suppress its own noise, and each pixel's summed spectrum is correlated against it along the energy axis and the position of the correlation maximum is that pixel's offset in detector channels. Every event matrix data is then shifted by the offset of the pixel it belongs to.

This type of offset correction was only needed for one data acquisitions. For the results shown in chapter 5, the offset was not present.

4.3.2 PINEM sidebands used for energy axis calibration and pulse detection

Electron crossing an excited optical cavity undergo an exchange process where they either lose or gain a photon to evanescent electromagnetic field of the cavity. This

phenomena is known as Photon Induced Near field Electron Microscopy or PINEM for short.



(A) (B)

FIGURE 4.4: Two set of measurements from PINEM sidebands. (a) Energy calibration and axis offset correction using the equal splitting of the PINEM sidebands. The spike at 1.8eV is a detector hot channel. (b) The pulse detection using intensity variation of PINEM peaks with time. The rise determines the t_{on} and the fall denotes the t_{off}

Due to the loss and gain, this stimulated signal appears in the EELS spectrum as two sidebands displaced from the zero-loss peak by multiples of $\hbar\omega$ [55]. For the case of 550nm laser pulse the PINEM sidebands must be at 2.254 eV. Key features of PINEM used here are the equal splitting of the gain and the loss (see fig. 4.4a) sideband which are used to calibrate the energy axis offset and set the dispersion per pixel of the EELS signal. Moreover, the fact that PINEM only happens in presence of an electromagnetic field is used to determine the time at which the laser pulse turns on. The PINEM intensity summed over all spatial pixels and cropped at the sideband energy is fitted with a twelth order polynomial. The rise and decay are detected by the extrema of its first order derivative. The order was chosen empirically for a good fitting of the raw data which follows the rise and fall of the signal (see fig. 4.4b).

The measured pulse width of $27-29$ ns (tab. 5.1) is slightly longer than the 25 ns imposed in the simulation (tab. A.2). Every fit in chapter 5 utilizes the t_{on} derived from figure 4.4b as a starting index for the laser pulse.

4.3.3 Rebin and cropping to improve counting statistics

The selected region of interest is cropped in the EELS spectrum. In this work, we use the aluminium bulk plasmon which is centered at about 15.5 eV (see fig. 2.3c). We crop the energy axis from 12.0 eV with 256 bins i.e. upto 18.0 eV. Similarly the time window is cropped from $0.8\,t_{\text{on}}$ with 2816 bins to the end of the useful record, exact range of the cropped signal is given in table 4.1. The length of the bins are chosen

as a multiple of two for rebinning. Rebinning adds statistics of the subsequent bins to increase the counting statistic and increase the number of counts per channel.

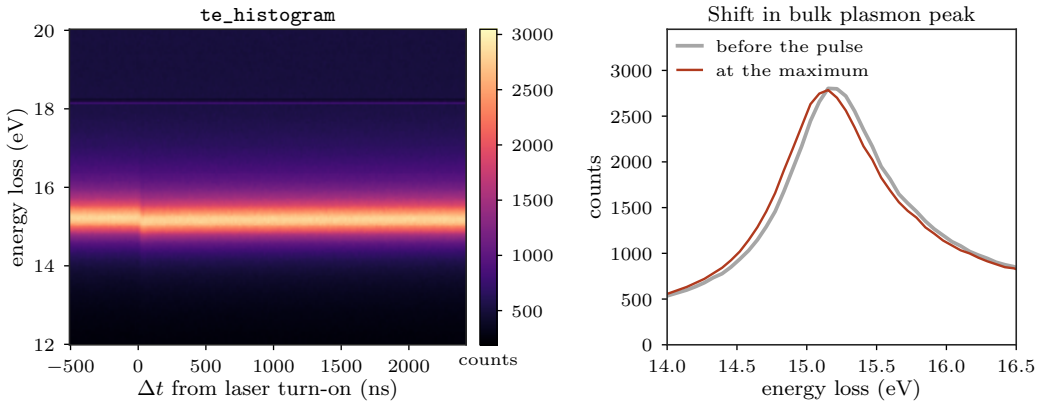


Figure 4.5: Left: The `te_histogram` showing the peak shift on heating at t = 0 . The faint horizontal detector artefact is visible at 18.0 eV. Right: the same data averaged over the pre-pulse bins and over the bins around the maximum. The separation between the two curves is due to temperature shift.

Table 4.1: Step sizes of the final fitting cube and the windows it covers.

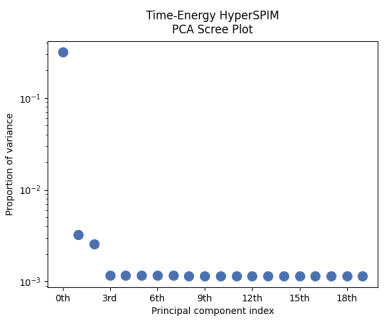
Axis	window	step	rebin factor	size
scan x, y	1900 nm field	59.4 nm	8	32×32
time	−508.06 – 2307.94 ns	2.083 ns	2	1408
energy	11.99 – 20.09 eV	63.3 meV	2	128

The time step is much shorter than the pulse width so the transient can be resolved properly. The rebinned scan step is in the order of the bridge width (tab. 3.1) so the details across the bridge is lost but the details along the bridge remain available. The resolution in this work is limited by the counting statistics rather than the method. A longer acquisition or a brighter electron source can resolve these issues with the acquisition limits (sec. 4.2).

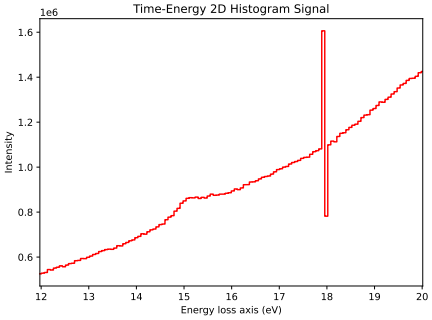
4.3.4 Masking and denoising with PCA

The data collected is very noisy but forms a repeating feature of the bulk plasmon whereas noise does not follow any repeating pattern. For a reliable Lorentzian fit and to determine the peak reliably, we need to extract the underlying features of the EELS spectrum.

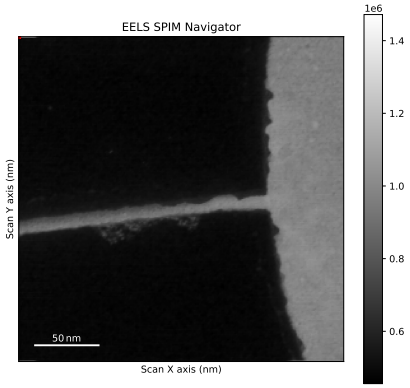
Principal Component Analysis (PCA) [37] decomposes the spectral data into components out of which a few leading components are used to recontruct the EELS spectrum for all (s_x, s_y, t) pixels. Figure 4.6a shows the first three components carry



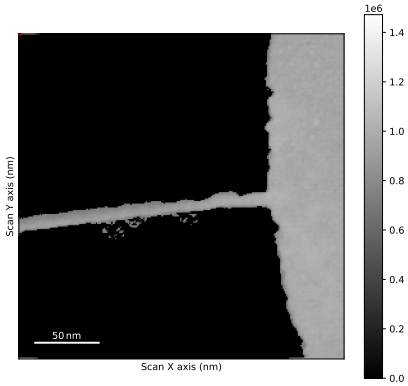
(A) Principal component decomposition.



(B) Pre-PCA minus post-PCA EELS spectrum.



(C) Before masking.



(D) After masking.

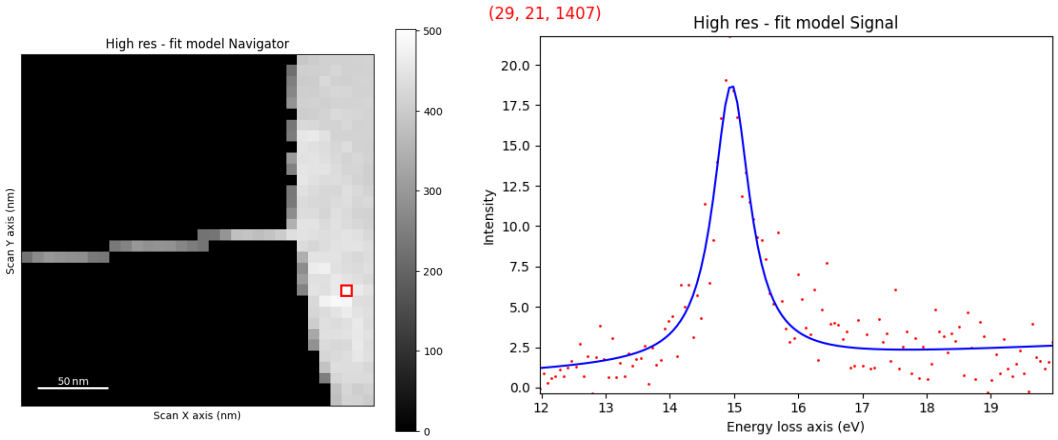
FIGURE 4.6: Masking and denoising. (a) Explained-variance spectrum of the decomposition of the binned cube. (b) pre-PCA minus post-PCA EELS spectrum. (c) The spectrum image with no mask and (d) with the aluminium mask applied.

most of the information, but truncating the reconstruction too hard biases the per-pixel centre towards the mean of the map. The recontructed EELS spectrum preserves the important feature of the spectrum while supressing the noise. Figure 4.6b shows pre-PCA minus post-PCA EELS spectrum where the underlying features were removed before Lorentzian fitting preserving the bulk plasmon feature . PCA requires a larger datasest to identify the principal compoments hence it is performed before masking the SiN membrane to have more signal for a reliable recontruction.

The SiN membrane is masked because a Lorentzian fit of a nitride membrane would return invalid values which when converted to temperature maps appear as atrefacts. Pixels whose time-integrated intensity in the aluminium bulk plasmon window falls below 0.25 of the maximum are excluded, which removes the nitride membrane. Along with this, it also decreases the compute so masking it becomes the most natural choice.

4.3.5 Lorentzian fit per-pixel-per-time bin

Starting from the cropped and rebinned `te_hyperspim`, the EELS spectrum is fitted with a Lorentzian curve with a linearly varying background. Figure 4.7 shows one such example. The centre of the Lorentzian is the resonance energy of the bulk plasmon and the instrument broadening is small against the intrinsic plasmon width hence no deconvolution is applied.



(a) (b)

Figure 4.7: Lorentzian fitting of the bulk plasmon. (a) A region on the tip pad. (b) the bulk plasmon peak extracted from the EELS spectrum of the corresponsing pixel at a given time.

The fit size is $286 \times 1408 = 402\,688$, and figure 4.7b shows the output of the multifit. The fit is performed with bounds on the centre and FWHM of the Lorentzian. The centre is bounded to $\pm 20\,\%$ and the width to $\pm 80\,\%$ of the summed-spectrum values, so a low-count fit cannot migrate away from the plasmon peak, and iteration follows a serpentine path through the navigation axes so that each fit starts from the converged parameters of its neighbour. Masked pixels are skipped and discarded, improving the runtime from a few hours to an hour and reducing the fit size from 32×32 to 286

This stage output a fit model with $E_p(s_x, s_y, t)$: a plasmon energy for every place and every moment, saved as a HyperSpy signal with the scan axes as navigation and time as the signal axis.

4.4 Plasmon energy to temperature conversion

In section 2.5 we established the physics of bulk plasmon energy and that it follows the free-electron density, the density falls as the lattice expands, and the peak red-shifts. In equation (2.3) we concluded that this relation can be inverted (see eq. 2.7)

to find the corresponding temperature change given by,

$$\frac{2}{3}\frac{\Delta E_p}{E_p(T_0)} = \alpha_0\,\Delta T + \frac{\alpha_1}{2}\,\Delta T^2 + \frac{\alpha_2}{3}\,\Delta T^3, \qquad \Delta E_p \equiv E_p(T_0) - E_p(T), \tag{4.1}$$

with the expansion coefficients of equation (2.6) taken from Chmielewski *et al.* [49, 50]. The sign convention makes a *decrease* in plasmon energy a positive ΔE_p and hence a positive temperature rise.

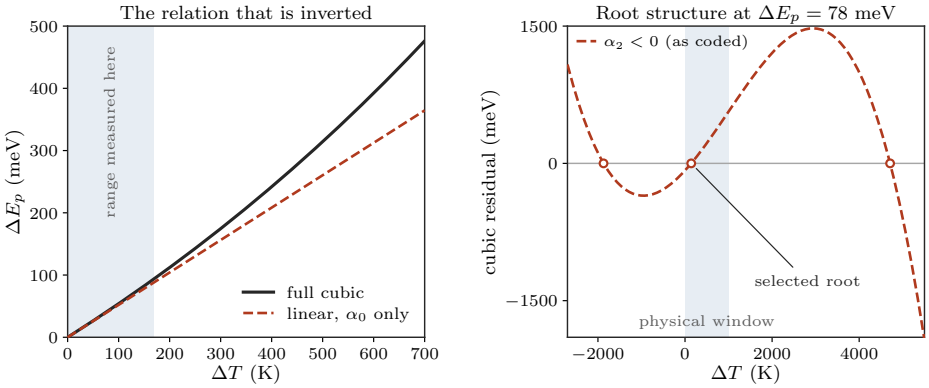


FIGURE 4.8: Left : ΔE_p vs ΔT plot with linear approximation at low ΔT assumption. Inverting linearly rather than with the full cubic overestimates ΔT, so the higher-order terms are kept. Right: Equation (4.1) is a cubic equation with three possible roots. Roots for a representative shift of ΔE_p =78 meV. In the code, the negative root at −1872 K is rejected as non-physical and out of the two real roots, the physical root is selected.

Equation (4.1) is solved for every pixel and every time bin, taking the smallest positive real root; the right panel of figure 4.8 shows the rule has a physical meaning. If no root below 1000 K exists, which happens in low-count pixels where noise dominates ΔE_p outside the physical range, the input shift is perturbed in steps of 0.5 % for up to fifty iterations, and the pixel is set to zero if that fails.

The ambient room temperature reference and baseline

The signal before the pulse is assumed to represent the ambient temperature bulk plasmon or each pixel. $E_p(T_0)$ is therefore established per pixel from the first ninety time bins, all of which precede the pulse. Their mean is calculated, and the largest per-pixel standard deviation is added to it. It biases the reference upward so that ΔE_p comes out positive almost everywhere and the cubic equation always has a physical root. This was done due to high noise in the temperature profile which leads to a negative ΔE_p value, giving inconsistent roots. As a consequence, we observe this nudge in the mean as a constant offset in the temperature, usually about $50-60$ Kelvin per pixel. Every rise and decay model used in chapter 5 fits this offset with a constant baseline.

Temperature maps

The pixel-by-pixel conversion of the energy maps produces temperature maps in time. Figure 4.9 shows a freeze frame of such temperature map with a resolution decided by the rebinning factor of table 4.1.

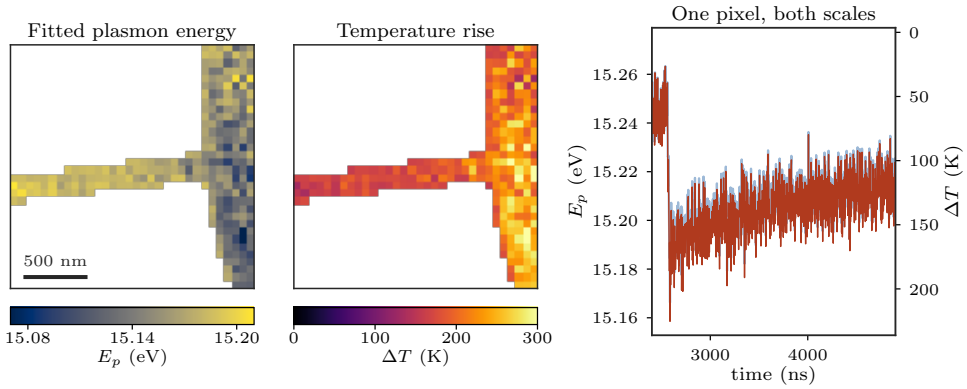


Figure 4.9: Conversion to temperature maps, time axis varying upto 2584 ns. Left: the fitted plasmon energy in every spatial pixel at a fixed time. Centre: the same pixels after equation (4.1) is solved in each of them; masked pixels are blank. Right: Time varying signal for one spatial pixel. The temperature axis inverted, so the traces lie on top of each other.

4.5 The decoupled rise and decay model

We saw in chapter 1, the Fourier's heat law (eq. 1.5) and discussed the analytical solutions for the case of a uniform thin film. In this case, we model the thermal rise during the laser pulse as an exponential with it's characteristic rise time. The modified Fourier's law gives,

$$\Delta T(t) = A\left[\exp\left(-\frac{t-t_{\text{on}}}{\tau_{\text{decay}}}\right) - \exp\left(-\frac{t-t_{\text{on}}}{\tau_{\text{rise}}}\right)\right] + C. \tag{4.2}$$

The rise and the decay exponentials are coupled together. Fitting the coupled heat transfer model shows a skewed measurement in the τ_{decay}. Alternatively, a decoupled model is investigated where the rising and falling exponentials have *independent* amplitudes. The decoupled temperature decay + rise model given by,

$$\Delta T(t) = \begin{cases} A_1 - A_2 + C, & t < t_{\text{on}}, \\ A_1 \exp\left(-\dfrac{t-t_{\text{on}}}{\tau_{\text{decay}}}\right) - A_2 \exp\left(-\dfrac{t-t_{\text{on}}}{\tau_{\text{rise}}}\right) + C, & t \geq t_{\text{on}}, \end{cases} \tag{4.3}$$

In chapter 1 we established that for the case of an uniform film, only the decaying exponential modes survive over a long time as the shorter eigen modes of the solution

die quickly. At a shorter time scale as measured here, we expect contributions from other eigenmodes as well. Additional exponential terms are considered to construct the decoupled exponential model. This model is observed to fit the rise and temperature peak better than the coupled model. Figure 4.10 shows the difference in τ_{decay} between the two models.

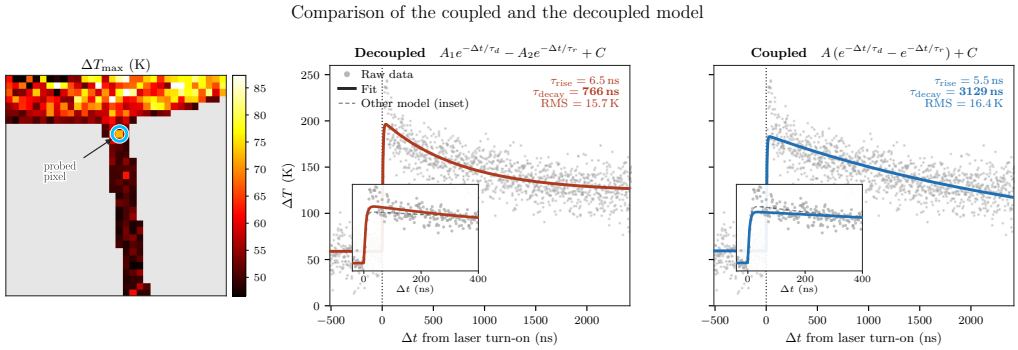


FIGURE 4.10: Comparison between the decoupled and the coupled heat rise+decay model. The peak of heat rise is modelled adequately by decoupled model.

The heat rise is due to an active laser pump whereas the heat decay is governed by the Fourier's heat law. The amplitudes do not have to be the same in this case. The parameters have an interpretation. The parameters and there meaning is tabulated in table 4.2.

TABLE 4.2: The parameters used in equation (4.3) and the quantities derived from them.

Symbol	Meaning	Role in the comparison
A_1, A_2	decay and rise amplitudes	fitted independently
τ_{rise}	rise time constant	time constant of the rising exponential only; set by the local heat capacity and the source
τ_{decay}	cooling time constant	time constant of the decay exponential, set by the thermal conductance.
C	long-time asymptote	not the pre-pulse level, which is $A_1 - A_2 + C$
ΔT_{max}	peak rise above C	scales with absorbed power.
$t_{\text{rise}\to\text{peak}}$	$t_{\text{peak}} - t_{\text{on}}$	transport time from the pad to the probe point

An important clarification is to be made between τ_{rise} and $t_{\text{rise}\to\text{peak}}$.

τ_{rise} is the exponential time constant of the rising exponential whereas $t_{\text{rise}\to\text{peak}}$ is the absolute time it takes for the region to peak the maximum of temperature. This qualitatively signifies the time delay of heat propagation.

Differentiating equation (4.3) gives,

$$t_{\text{peak}} = t_{\text{on}} + \ln\left(\frac{A_1\tau_{\text{rise}}}{A_2\tau_{\text{decay}}}\right)\frac{\tau_{\text{rise}}\,\tau_{\text{decay}}}{\tau_{\text{rise}} - \tau_{\text{decay}}}. \tag{4.4}$$

Under the coupled model $A_1/A_2 = 1$, and $t_{\text{rise}\to\text{peak}}$ is then fixed by the τ_{rise} and τ_{decay} .

4.6 Conclusions

This chapter elaborated the data analysis and data processing pipeline developed in this internship (fig. 4.1). The data structure and the python libraries used were discussed briefly (sec. 4.2) along with the acquisition limits during the experiments (sec. 4.2). The data processing and the analysis pipline was described in detail with elaborative steps, highlighting the limitation of this technique - the counting statistics or the beam current (sec. 4.3). Finally, this chapter provided details of the decoupled rise and decay model used to construct the temperature maps (sec. 4.5).

Chapter 5

Results and comparison with simulation

The final experiments were performed on two nanobridges of nominal width 50 nm and 150 nm, named as datasets `02` and `04`. The final spatial and temporal resolutions for each data set is 59.4 nm and 2.08 ns after rebinning (tab. 4.1). These temperature maps are fitted with the decoupled rise-and-decay model of section 4.5 (eq. 4.3) and the relevant fit parameters are extracted for each pixel. This chapter discusses the results obtained from the model fitting, temperature maps and compare the experimental results with the COMSOL - FEM simulations of chapter 3.

5.1 The measured datasets

The experiment was performed on the second generation design with elliptical tip pads. The thickness aluminium layer was 24.8 nm (app. C.1, tab. C.1). The bridge widths were measured from HAADF images in appendix C.2 and are tabulated in table 3.1 of chapter 3. As the ChromaTEM has a limited field of view, the bridge length visible in the measured data was analysed as it has to be compared with the COMSOL simulations in section 5.4 and needs a distance-from-pad coordinate. That measurement is set out in appendix C.3. Figure 5.1 presents the two datasets used for this chapter.

TABLE 5.1: The two measured bridges measured in app. C.3. The pulse width is determined per acquisition as in section 4.3.2.

Dataset	Design width	Measured width (nm)	Visible length (nm)	Pulse width (ns)
`02`	50 nm	59.9 ± 1.6	1458 ± 30	28.65
`04`	150 nm	174.4 ± 2.1	1527 ± 32	27.86

Each dataset records a slightly different pulse width time by the pulse width detection algorithm. To preserve the counting statistics, the 50nm bridge is highly rebinned (see sec. 4.3.3), giving just one pixel along the bridge. The other dataset

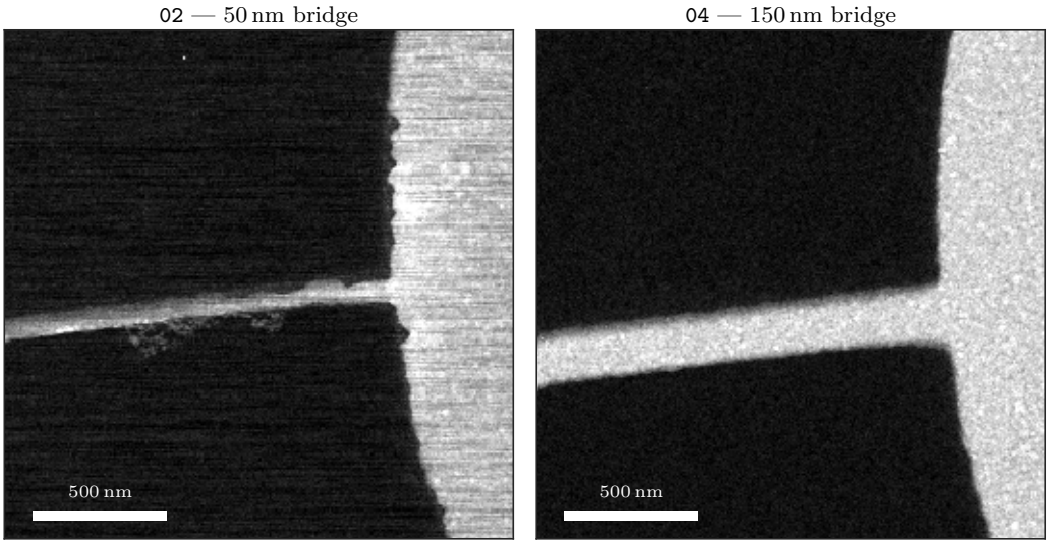


Figure 5.1: HAADF images of the two bridges. Left: nanobridge with nominal width 50nm. Right: nanobridge with nominal width 150nm

is four pixels wide but does not carry any significant information across the bridge. Therefore the analysis performed in this work is along the bridge.

5.2 Temperature evolution at nanometre scale

The temperature maps extracted in section 4.4 for the 04 dataset show a clear evolution of the heat dissipation in the nanostructure in figure 5.2. Heat appears first on the illuminated tip pad and reaches the far end of the field about 30 ns later (see fig. 5.2). The pad stays the hottest point throughout and the bridge follows the evolution at a reduced amplitude. This means that we are pumping the heat on the pad and probing how the system relaxes a certain distance away.

At the last frame in the record the structure is still measurably above baseline. For a limited acquisition window, the data for complete cooldown is not available. It was shown by Castioni *et al.* [7] that the uniform thin film cools down completely before the next pulse which is 100μs apart (tab. A.2).

A single freeze frame is not a quantitative measure of the heat transport but provides a qualitative picture of the evolution. For a quantitative picture, we fit the heat evolution curves with the decoupled heat evolution model for each spatial pixel.

The field of view covers the tip pad and the first 1.5 µm of the bridge (tab. 5.1) whereas the total bridge length spans 5 µm (tab. A.2) therefore the heat-sink pad is outside the field of view. Distances quoted from here on are measured from the pad edge along the bridge.

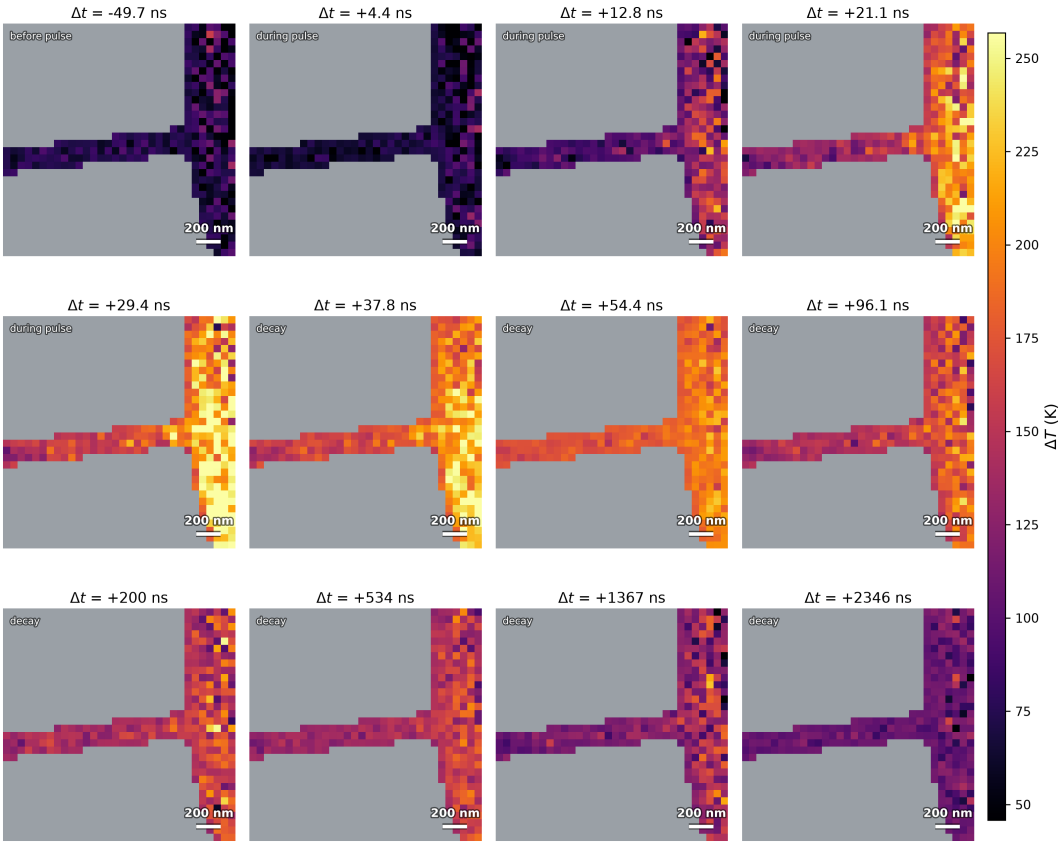


FIGURE 5.2: Temperature evolution heat maps of the 150nm nanobridges.

5.3 Extracted fit parameter maps

For each dataset, the temperature model fits the temperature evolution. Figure 5.3 shows the model fitting for the 150nm nanobridge dataset. The rise and decay decoupled model fits the experimental data along the bridge for each dataset which shows the reliability of the decoupled model.

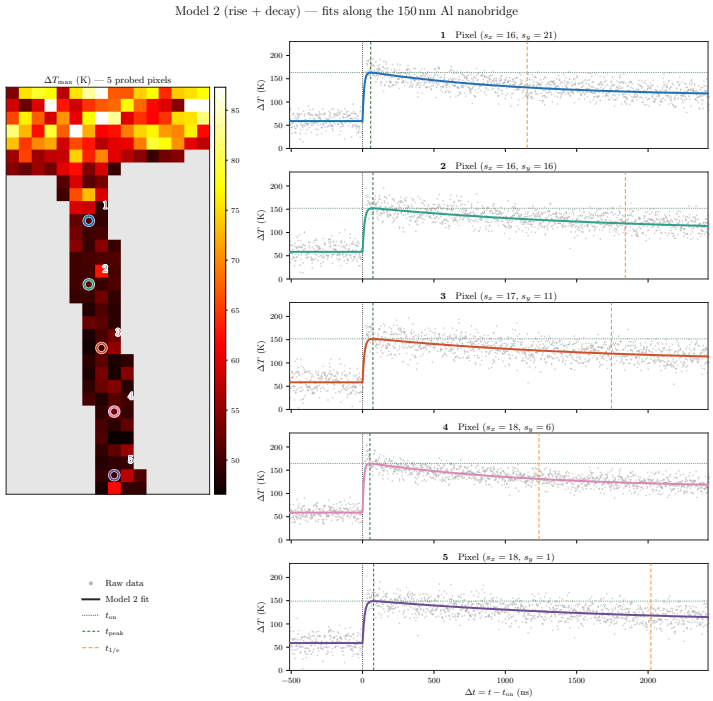
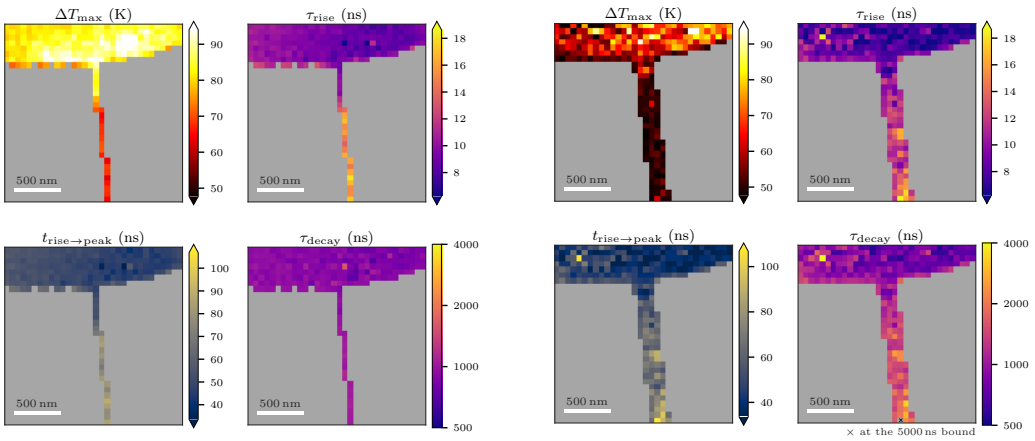


Figure 5.3: Decoupled model fit at five points along the 150 nm bridge. The fit parameter maps are shown in the next section.

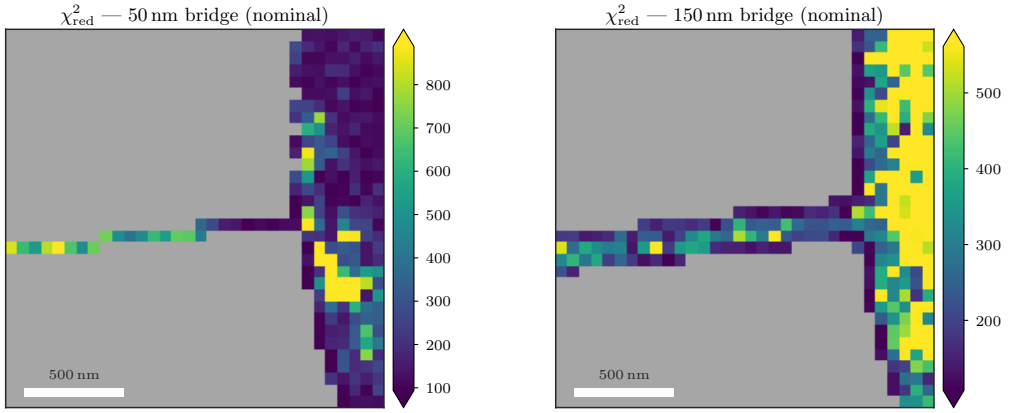
The fit parameters explained in table 4.2 are extracted for each pixel for the two datasets and presented as spatial maps in figure 5.4. The figure shows that the $\tau_{decay}, \tau_{rise}$, and $t_{rise \to peak}$ all grow with distance from the tip pad. $\Delta T_{\max}$ falls from the tip pad towards the bridge section as the laser heating occurs on the pad.



(A) (B)

FIGURE 5.4: Extracted parameter maps (a) shows nanobridge of nominal width 50nm and (b) shows the nanobridge of nominal width 150nm.

The fit quality across every pixel is measured by χ^2 parameter maps of the fit maps, shown in figure 5.5. It was observed that near the pad the decay is fastest and best resolved for the 50nm bridge. The EELS data is noisy in the bridge section leading to a higher χ^2 in the bridge region of figure 5.5a and in the figure 5.5b, for the 150nm nanobridge we observe a higher spread of the EELS data in the pad area leading to a higher χ^2 value. It is better resolved in the bridge region.



(A) (B)

FIGURE 5.5: The χ^2 maps of the two datasets.

It must be noted from figure 5.3 that the heat dissipation curves have not been analysed for a longer time of temperature decay. This was due to the limited size of the datasets. Each dataset requires a preprocessing time width window. Choosing a

larger window increases the file size, which was out of the scope of the algorithm developed in this work.

5.4 COMSOL simulations and comparisons

The finite-element runs of chapter 3 were fitted with the same decoupled equation (4.3) follwoing the same procedure, and the simulations and experimental measurements are compared in this section. The simulations were performed on a model bridge of width 200 nm. Figure 5.6 shows the decoupled model fits on the COMSOL simulations. The curves with decreating T_{peak} represent probes away from the tip pad as shown in figure 5.3.

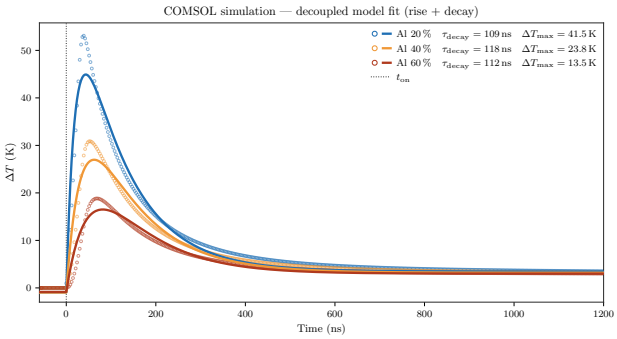


Figure 5.6: Decoupled model fit shows a mismatch in the fit at the peak.

Figure 5.6 shows a mismatch at the peak of the temperature curves, where the model underestimate the fit peak.

The laser power in the simulations was tuned to match the ΔT_{max} of the experiment at the beginning of the bridge. As the laser heating in COMSOL was modelled with `laser beam heating` which does not model the light matter interaction, the laser power had to be manually tuned.

The fit parameters are extracted along the bridge with the same field of view as the experiments and reported in figure 5.7 with $0\mu m$ as the point near the tip pad and $1.5\mu m$ as the far end of the field of view of the nanobridge.

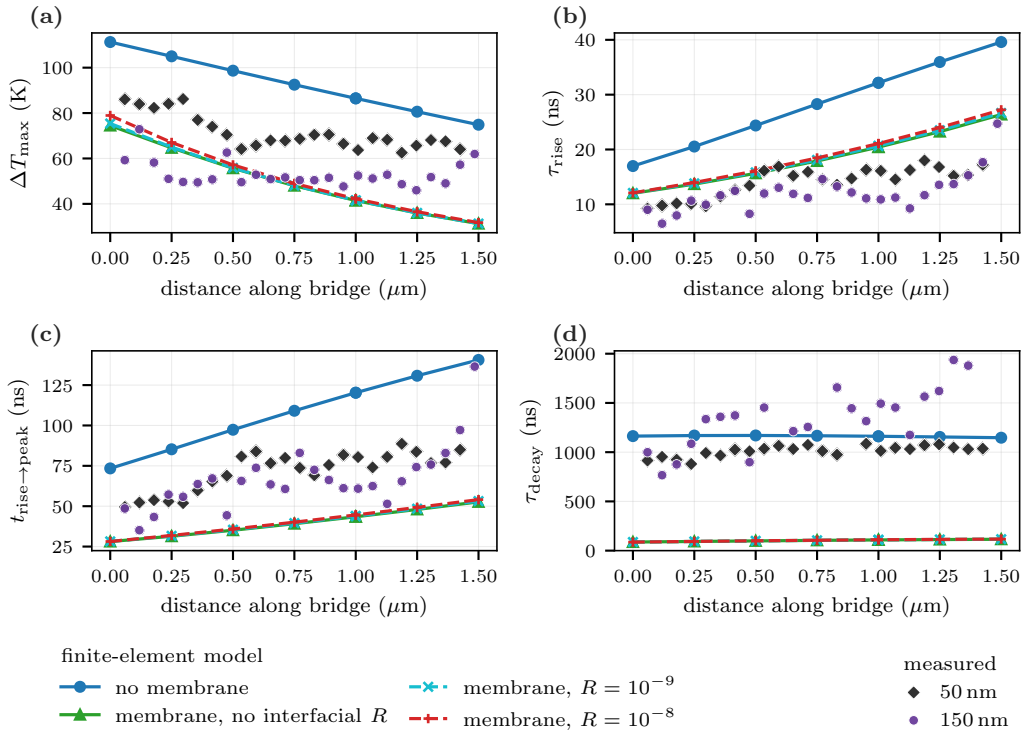


FIGURE 5.7: The comparison plot of the COMSOL simulations and the experimental datasets of nanobridges with nominal width 50nm and 150nm.

From the figure 5.7 we can notice,

(a) The ΔT_{max} falls by $\sim 20K$ for the case of 50nm bridge whereas it remains at $\sim 50K$ for the case of 150nm bridge slightly decreasing as we move away from the tip pad. It was only adjusted in simulations to match the initial few values, the trend is an outcome of the Fourier heat transfer model.

(b) The τ_{rise} increases from $\sim 10ns$ to $\sim 18ns$ along the bridge from the tip pad to the sink pad which is consistent with the simulations. As we move away from the heat source, the rise time increases due to the distance from the heat source.

(c) The $t_{rise\to peak}$ which is the absolute time taken to reach the maximum of temperature, closely follows the trend of τ_{rise}. We saw in equation (4.4) that they are linked to each other by the amplitude ratio A_1/A_2.

(d) The τ_{decay} remains flat with distance along the 50nm bridge but increases significantly for the 150nm bridge. It is expected to remain flat in diffusive transport where the device geometry does not affect the propagation of carriers.

To further investigate the role of SiN as a potential heat dissipation channel, the interaction of aluminium bridges with the SiN layer was simulated by varying the interfacial resistance between the two materials (see sec. 1.5). Table 5.2 shows various simulation conditions.

The configuration with no SiN simulates the case of zero dissipation to the SiN

Table 5.2: Simulation conditions and corresponding SiN thermal resistance values.

Condition	Substrate	Interfacial Thermal Resistance(ITR) (R)
1	No SiN	—
2	SiN	0 m^2KW^{-1}
3	SiN	10^{-8} m^2KW^{-1}
4	SiN	10^{-9} m^2KW^{-1}

membrane and assumes all the heat to pass through the nanobridge region imitating the case of infinite ITR. The configuration with zero ITR simulates the SiN as a potential heat sink with unresisted heat flow from the aluminium to the membrane, whereas the two cases of with finite resistance shows the two extreme cases of ITR between aluminium and SiN membrane. The range of thermal interface values were taken from P.Hopkins *et al.* [56]. The two cases of "no SiN" with infinite ITR and "SiN" with zero ITR represent the two extreme cases of heat dissipation to the membrane.

For the case of infinite ITR, the τ_{rise} and $t_{rise \to peak}$ are overestimated. This is highly counter-intuitive as it was expected that the presence of SiN would increase the rise time as it takes more time to cause a temperature rise, given an alternate dissipation channel. ΔT_{max} as expected is higher as all the heat flows through the aluminium, increasing the peak temperature. The relative decrease ramains consistant with the case of zero ITR. The decay time τ_{decay} in the case of zero ITR is less than the infinite ITR case as the SiN membrane is expected to provide an alternative dissipation channel pulling the heat away from the bridge.

We observe from figure 5.7 that the presence finite ITR [56, 57] does not have a significant effect on coupling of the aluminium and the SiN membrane suggesting, an underestimation in the ITR values.

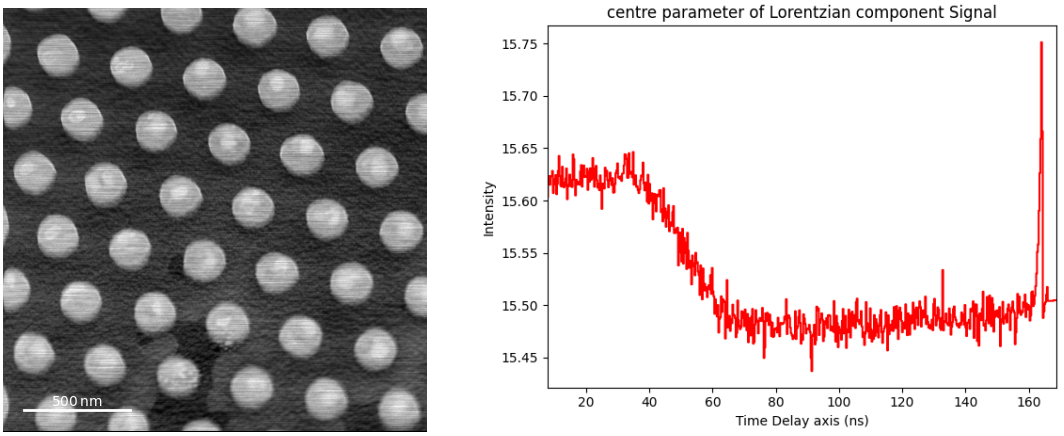
The results show that we are able to observe the temporal delay of heat within the bridge, regardless of its size. Naturally, the size of the bridge does not influence the dynamics of heat flow propagation since the electrons carrying the heat, have a mean free path much smaller than the size of these structures [58] (see sec. 1.1 and sec. 1.4). Nevertheless the ability to observe these effects in a nanostructure demonstrates the robustness of this approach. Measurement of these effects in a 50nm wide bridge indicates that studying these same effects in a material where the heat is instead carried by phonons (typically a semiconductor) should also be possible. In this case, the heat propagation would be affected as phonons have a longer mean free path compared to the electrons.

The technique was established to study model nanobridge but it also applicable for other nanostructures observed in this work. Thermoplasmonic structure are a good candidate to study nanothermometry hence this technique was applied on a nanopillar thermoplasmonic structure.

5.5 Thermoplasmonic structures

Thermoplasmonic structures convert light into localised heat by absorbing a characteristic wavelength and resonate to produce heat. In this case, aluminium deposited upon SiO_2 nanopillars etched on a solid substrate were used as thermoplasmonic structures provided by a collaborator at DTU [59]. The pillars and the bulk plasmon peak shift in the space between the pillars is illustrated in figure 5.8. Figure 5.9 maps the heat flow upon laser illumination.

The pillars are roughly 200 nm in diameter (see fig. 5.8a). Illuminated by a laser at 550 nm (2.25 eV),whereas the pillars are resonant at 2.5 eV as reported by Kadkhodazadeh *et al.* [59]. Although the laser energy is off resonant, the pillars absorb the laser energy which leaves through the base into the substrate (see fig. 5.8a). It must be noted that the astigmatic laser (see sec. 2.4) also heats the aluminium at the base of the pillars. It can not be definitively said that the melting observed in figure 5.8a is only because the plasmonic structure. A combination of heat absorption by the aluminium and the nanopillars create thermal spots near the base of the pillars visible as the thin aluminium layer melting from the base of the pillars.



(A) (B)

FIGURE 5.8: Thermoplasmonic structure studied using STEM-EELS. (a) A top-view HAADF image of the plasmonic array shows the aluminium thin film melting from the base of pillars. (b) bulk plasmon peak shifting over time upon heating

One question that remains, is if we are able to map how the optical excitation turn into heat, and how the heat flows around this plasmonic structure, and how much time they need to thermalise and evacuate the heat.

This measurement was performed to test the variability of the technique and shows a promising scope to investigate proper nanostructures with interesting applications.

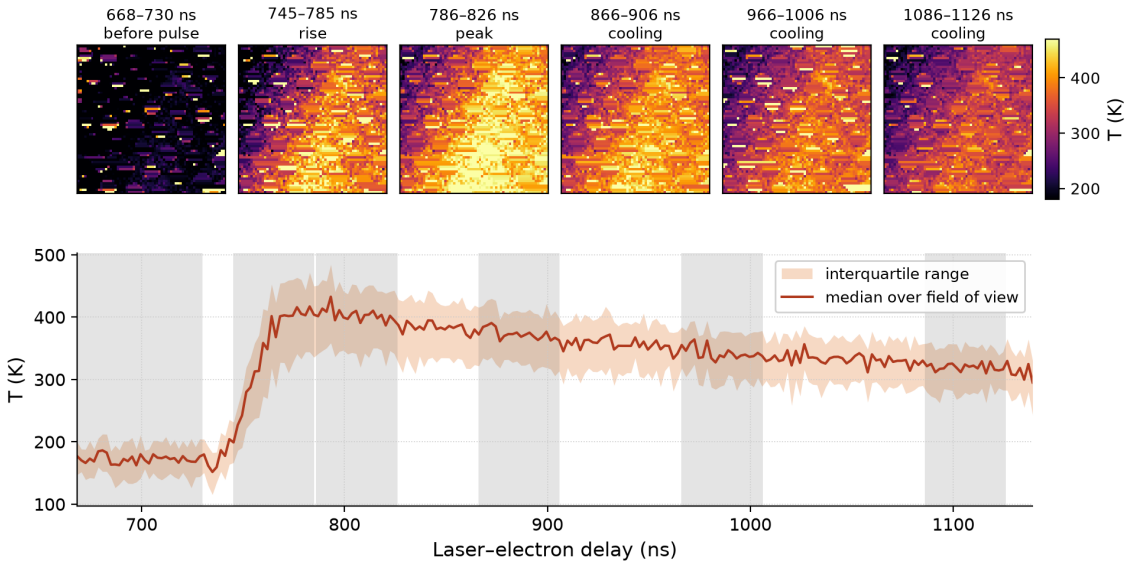


Figure 5.9: Heat arriving upon laser illumination propagating through the structure. The six panels are the temperature maps of the same field of view, each summed over the delay window printed above it: one before the pulse, one during the rise, one at the peak, and three during the decay. The plot below is the median temperature over this field of view against laser-electron delay, with the interquartile range shaded and the six windows marked. The rise occurs in tens of nanoseconds and the decay takes several hundreds of nanoseconds.

5.6 Conclusions

This chapter presented the results obtained on dynamic temperature measurement in a metallic nanostructure with nominal width 50 nm and 150 nm, each giving a temperature map at 59.4 nm and 2.08 ns (tab. 4.1) of spatial and temporal resolution respectively. A decoupled rise-and-decay model was fitted at every spatial pixel. The fit parameters carry the important transport information.

Similar results were obtained in the two datasets of two different bridge widths. $\Delta T_{\max}$ is largest on the illuminated tip pad and falls along the bridge, while τ_{rise} and $t_{\text{rise}\to\text{peak}}$ grow with distance from the pad. The χ^2 maps place the best fits near the pad, where the acquired data is less noisy and worst at the noisy bridge data.

Against the finite-element simulations their is a partial agreement in the decoupled model. The measured fall in $\Delta T_{\max}$ and the rise of τ_{rise} along the bridge are both reproduced, and $t_{\text{rise}\to\text{peak}}$ is between the membrane-free and perfectly-coupled extremes. For the τ_{decay}, the experimental data does not record the long time response of the temperature curves, making it difficult to predict the behaviour. ITR changes little across the range tested, so the membrane matters as a parallel path of heat dissipation and the coupling needs to analysed further. The actual interface between the SiN and the aluminium might not be perfect and might contain organic contaminents from the e-beam resist, causing an increase in the ITR. The agreement with simulations show that the underlying physics is well understood, but needs additional modelling.

The measurement is not specific to bridges and one dataset only. On the aluminium-coated nanopillars the same temperature mapping resolves a rise of tens of nanoseconds and a decay of several hundred, over the field of view. The same experiment was repeated on the first generation square tip pad nanobridges (see fig. 3.3a) which give a similar result.

Chapter 6

Conclusion and outlook

This work extends the time-resolved nanothermometry technique from a uniform thin film to a confined nanostructure and maps the heat transport along a metallic nanobridge with nanometre and nanosecond resolution simultaneously. Aluminium nanobridges of nominal width 50nm and 150nm with a film thickness of 24.8nm were patterned by electron-beam lithography on a 15 nm silicon nitride membrane, heated by a focused laser pulse on a tip pad at one end, and their temperature is extracted from the shift of the aluminium bulk plasmon in the time resolved electron energy loss spectra (Sec. 2.5). The resulting temperature maps resolve 59.4 nm spatially and 2.08ns temporally (Sec. 5.2), on a structure well below the diffraction limit of any optical thermometry technique together with a timescale relevant for the study of heat propagation at these scales. The combination of both temporal and spatial capabilities at the same time has, to our knowledge, never been reached before. (Sec. 1.6).

Finite element simulations were performed to simulate the heat dynamics in the nanobridge using COMSOL with the `heat transfer in solids` module (Chap. 3). COMSOL assisted nanobridges were designed and investigated for efficient thermal transport through the nanobridge region. Fabrication process for the nanobridges was established (Sec. 3.4) and dose optimisation studies were performed to calibrate the dose specific to electron beam lithography on a thin membrane (Sec. 3.4.1).

A robust data processing pipeline was established, to convert the raw data into time resolved temperature maps (Sec. 5.2). The extracted time resolved temperature maps were modelled using a decoupled exponential model (Eq. 4.3) which showed a good fit for the temperature window available (Sec. 4.5). Fit parameters extracted from the model were reported for each spatial pixel (Sec. 5.3) and compared with the finite element simulations (Sec. 5.4). The temperature rise is observed to be the largest in the illuminated area and falls with distance from the pad. The rise time and the time to reach the maximum temperature after the laser pulse, grow with the distance from the pad rising from 50ns near the tip pad region to $\sim 75ns$, $1.5\mu m$ away from the tip pad along the bridge (Sec. 5.4). The decay time does not vary significantly, as expected from a diffusive transport. The mean free path of the electrons carrying heat is far shorter than the dimensions of the bridges used.

Role of the SiN membrane as a potential heat dissipation channel was investigated. The interfacial thermal resistance (ITR) between the aluminium bridges and the SiN membrane was investigated between two extremes - zero ITR and infinite ITR. For the case of finite resistance, no significant change in the dynamics is observed, requiring further investigation.

This work can be extended to investigate the role of ITR in the heat dynamic and more generally to study the ITR (also known as Kapitza resistance) between two materials especially for confined nanostructure were typical pump-probe optical techniques fail to isolate the object of interest. The technique can be use to study heat transport in microelectronic systems or other complicated designs (Sec. 5.5). Advancements in the present microscope enabes it probe more fundamental heat transport physics, such as the optical excitation phenomena responsible for heat generation with potential access to the phonon dynamic on confined systems.

Appendices

Appendix A

The finite-element model

All simulations used COMSOL Multiphysics 6.4, *Heat Transfer in Solids*, three-dimensional model.

A.1 The model file as configured

Table A.1: Provenance of the model file the parameters below were read from.

Item	Value
Model file	`Al bridge time delay - demo 1 .mph`
COMSOL version	Multiphysics `6.4.0.343`
Component	`comp1`, one component, three-dimensional
Interface	*Heat Transfer in Solids* (`ht`), transient
Initial temperature	293.15 K
Outer boundaries	*Thermal Insulation*

Table A.2: The `Parameters table` used in the model file.

Name	Expression	Value	Description
`absorbivity`	0.09	0.09	absorptivity
`bridge_h`	`pad_h`	5×10^{-8} m	bridge thickness
`bridge_l`	`5[um]`	5×10^{-6} m	bridge length
`bridge_w`	`200[nm]`	2×10^{-7} m	bridge width
`D`	`1.5[um]`	1.5×10^{-6} m	laser width
`duty_cycle`	`pulse_w/pulse_time`	3×10^{-4}	duty cycle
`laser_dev`	`D`	1.5×10^{-6} m	Gaussian width σ
`laser_loc`	0	0	spot centre
`p_pulse`	`10[mW]`	0.01 W	pulse power
`pad_h`	`50[nm]`	5×10^{-8} m	aluminium thickness
`pad_l`	`2[um]`	2×10^{-6} m	tip pad length
`pad_sink_l`	`20[um]`	2×10^{-5} m	sink pad length
`pad_sink_w`	`10[um]`	1×10^{-5} m	sink pad width
`pad_w`	`2[um]`	2×10^{-6} m	lamella dimensions
`pulse_time`	`100[us]`	1×10^{-4} s	pulse repetition time
`pulse_w`	`30[ns]`	3×10^{-8} s	pulse on time
`sim_end`	`pulse_w*100`	3×10^{-6} s	solved interval
`SiN_h`	`15[nm]`	1.5×10^{-8} m	nitride thickness
`SiN_l`	`SiN_w`	3×10^{-5} m	nitride domain length
`SiN_w`	`30[um]`	3×10^{-5} m	nitride domain width

Table A.3: Mesh element sizes, one free-tetrahedral operation per domain, all with a custom element size calibrated for general physics.

Free-tetrahedral operation	Domain	Maximum	Minimum
`SiN`	1	1 µm	500 nm
`pads`	2	3 µm	500 nm
`pad small`	3	500 nm	100 nm
`bridge`	4	`pad_h/2`, 25 nm	`pad_h/4`, 12.5 nm

A.2 Material properties

Table A.4: Material properties given to the solver. The nitride values are sourced from [60, 61].

	k (W m^{-1} K^{-1})	ρ (kg m^{-3})	c_p (J kg^{-1} K^{-1})
Aluminium	238	2700	900
Silicon nitride	3.0	3100	700

Appendix B

Fabrication

The process recipe and the final layout of the aluminium nanobridge arrays.

B.1 Process steps

Table B.1: Process flow for the aluminium nanobridge arrays, one row per recorded parameter, grouped by process step.

Step	Parameter	Value	Note
Substrate	Product	TED PELLA no. 21569-10	Purchased lamella; no in-house membrane fabrication.
	Frame	3.05 mm Si	
	Membrane	(15 ± 3) nm SiN	Nominal, as supplied.
	Window array	3×3, 100 µm windows	
	Window separation	350 µm	
	Mounting	Ag paste on a Si carrier	One edge of the array kept parallel to the wafer edge.
	Pre-inspection	Optical, all nine windows	For cleanliness and for the presence of all nine membranes.
Resist	Prebake	180 °C, 15 min	Carrier and pasted lamella together, before spinning.
	Resist	PMMA 950 A3	3 % solids in anisole.
	Spin speed	3000 rpm, 60 s	
	Spin ramp	2000 rpm s^{-1}	
	Resist thickness	150 nm expected	Never measured; value from the spin curve.

continued on the next page

Table B.1: *(continued)*

Step	Parameter	Value	Note
	Post-bake	180 °C, 4.5 min	
	Rejected variant	PMMA 950 A6	6 % solids; cracked the membrane during the post-bake.
Exposure	Tool	Zeiss Supra 55 VP	SEM with a lithography stage.
	Accelerating voltage	20 kV	
	Aperture	10 µm	Chosen to limit the beam current.
	Beam current	62.3 pA	
	Area dose	600 – 650 µC cm^{-2}	At the anchor layer; the per-layer fractions are printed in figure B.1.
	Focus	Per window	The lamella does not sit flat on the carrier, so focus is reset on each square.
Development	Developer	MIBK:IPA 1:3	
	Time	60 s	Continuous agitation by tweezers.
	Stop	Rinse in pure IPA	Immediate, to arrest development.
	Temperature	`RT`	Room temperature
Metallisation	Method	Thermal evaporation of Al	
	Target thickness	50 nm	
	Deposited thickness	24.8 nm	Run cut short by a technical failure.
	Chamber pressure	1.15×10^{-6} mbar	
	Source current	190 A	
	Deposition rate	3 Å s^{-1}	
Lift-off	Bath	Acetone, 50 °C, 10 min	First beaker of a two-beaker process.
	Flake removal	Tweezers	Larger liberated flakes fished out.

continued on the next page

TABLE B.1: *(continued)*

Step	Parameter	Value	Note
	Second stage	Acetone from a dropper	Flow directed along the bridge axis, never across it.
	Agitation	None	Ultrasound would destroy the membrane.

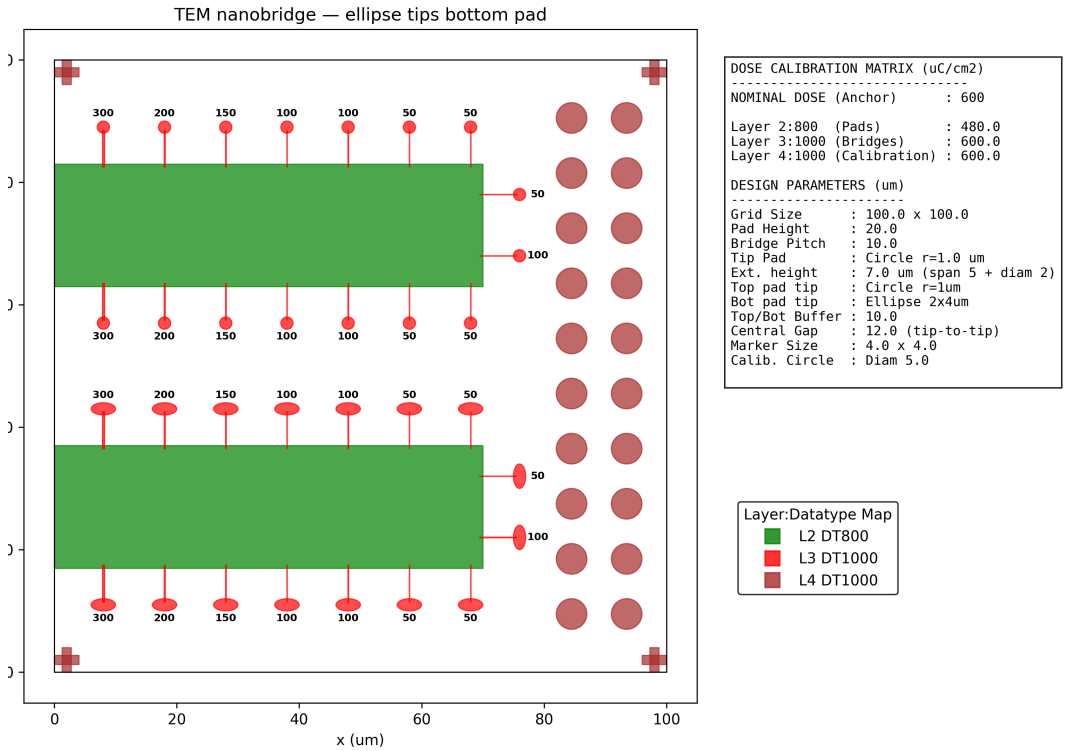


FIGURE B.1: The final layout as exported, with its own dose and design legend. Green is the sink pads, red the bridges and their tip pads, and dark red the calibration discs and the four corner markers; the numbers over each bridge are the drawn widths in nanometres.

Appendix C

Bridge dimensions

This appendix collects the two measured dimensions of the bridges: the thickness of the film, the width of each design, and the length visible inside the scan field.

C.1 Film thickness

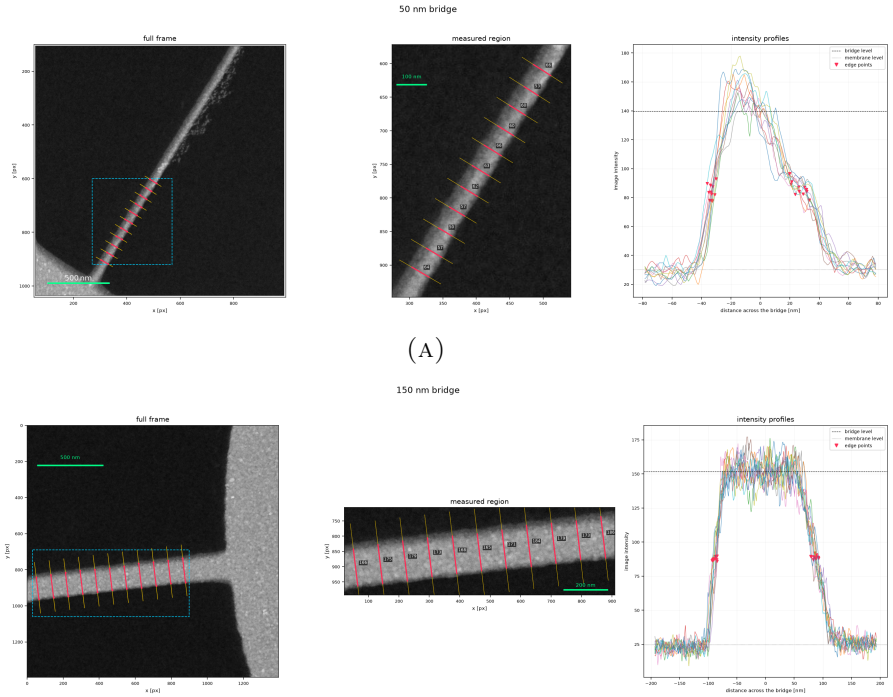
TABLE C.1: Out-of-plane dimensions of the bridge stack. No thickness metrology was performed on the finished bridges, so these are the specified and the deposited values.

Layer	Thickness (nm)	Remark
Silicon-nitride membrane	15 ± 3	grid specification
Evaporated aluminium film	24.8	targeted at 50 nm; the run was cut short by a technical failure

C.2 Bridge width

TABLE C.2: Bridge widths measured from the HAADF images, one row per design; the same measurement as table 3.1 of the main text. Each width is the mean of N transverse profiles with its combined standard uncertainty, and the variation is their standard deviation along the bridge.

Design	Pixel size (nm)	N	Measured width (nm)	Variation (nm)
50 nm	2.192 120	11	59.9 ± 1.6	5.2
150 nm	1.361 437	11	174.4 ± 2.1	6.9



(A)

(B)

Figure C.1: Width measurement: (a) the 50 nm design and (b) the 150 nm design. Each row shows the full frame with the measured region outlined, that region enlarged with the transverse profile lines, and the profiles themselves.

C.3 Bridge length

Table C.3: Visible bridge lengths for the 50 nm design (02) and the 150 nm design (04), measured along the fitted bridge axis from the tip-pad edge to the boundary of the scan field; both frames span a 1900 nm field. The subscan in the last row carries no pad edge and yields no length.

Dataset	Raster	Pixel size (nm)	Axis tilt (°)	Visible length (nm)
02	256×256	7.4219	-6.80	1458 ± 30
04	1024×1024	1.8555	-6.83	1527 ± 32
Not used: 185×185 subscan over 344 nm				

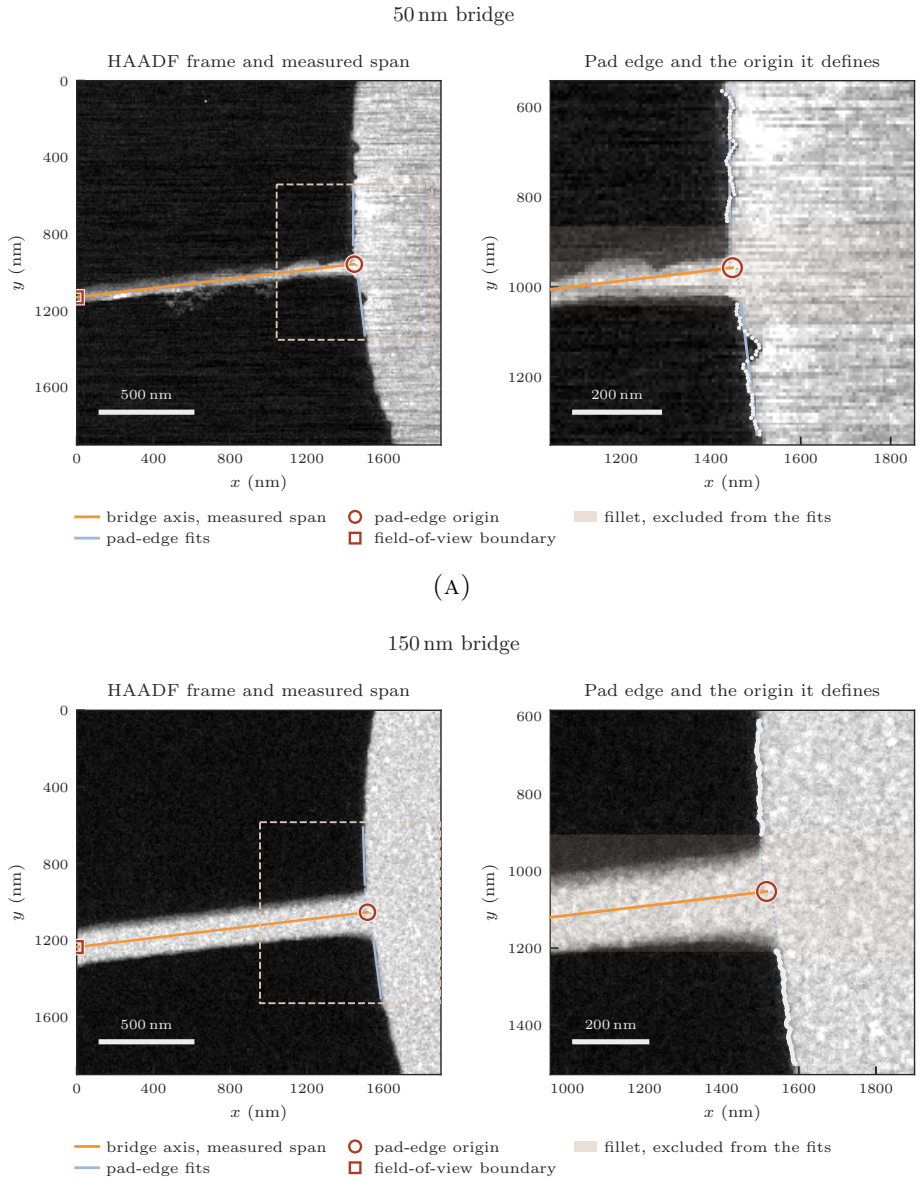


FIGURE C.2: Length measurement on (a) dataset `02`, the 50 nm design, and (b) dataset `04`, the 150 nm design. Left in each, the HAADF frame with the fitted bridge axis drawn over the measured span; right, the bridge-to-pad junction with the two pad-edge fits and the origin they define.

Appendix D

Acquisition conditions

D.1 Instrument state

TABLE D.1: Acquisition conditions for dataset `04`, from the instrument log of 1 July 2026.

Parameter	Value	Note
Microscope		
Instrument	ChromaTEM	modified Nion Hermes 200, cold FEG, monochromated
Accelerating voltage	100 kV	maximum transferable energy 5.2 eV, below the displacement threshold for Al
Convergence semi-angle	25 mrad	`25mrad8IRef_PP`
Monochromator slot	−0.325 mm	high-dispersion setting
Probe current	8 pA	
Spectrometer and detector		
Nominal dispersion	31.576 meV	per channel, from the acquisition metadata; 31.650 meV after the PINEM calibration of section 4.3.2
Energy origin	−6.3152 eV	channel zero before calibration
EELS aperture	2 mm	
Camera	Timepix3	event mode, threshold low, 1025 dispersive channels across four chips
Delay clock	0.2604 ns	one tick, = 1.5625 ns/6
Accumulation time	200 ms	
Excitation		
Source	pulsed dye laser	parabolic mirror, section 2.4
Wavelength	550 nm	$\hbar\omega = 2.2543$ eV
Power setpoint	41.9 µW	PMT2 reading
pulse width	25 ns	programmed in the laser module
Repetition rate	10 kHz	100 µs between pulses, the value assumed in the simulation
Scan		
Scan array	256 × 256 pixels	

Parameter	Value	Note
Pixel size	7.421 875 nm	field of view 1900 nm
Dwell time	128 µs	
Frames	200	
Data volume	310.2 GB	

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Code of conduct and transparency statement on the use of GenAI for KU Leuven-students (academic year 2025–2026)

Generative AI (GenAI) assistance tools can be used to generate various types of content, including text, images, code, video, music, or combinations thereof. Common examples of such tools include ChatGPT, Google Gemini, Microsoft Copilot, Midjourney, Claude.ai, Perplexity.ai, and DALL · E, among others.

This code of conduct is a tool that helps students to be transparent about the use of GenAI and fits within the university's principles on academic integrity.

Important guidelines and remarks

- **Sensitive or personal data:** Some GenAI tools protect your input and sensitive or personal data better than others. There is often no transparency on what the owners of the AI applications do with the data entered. Therefore, **do not enter sensitive or personal data in free GenAI tools.** More info about what sensitive and personal data are, is described in the three confidentiality levels of the KU Leuven data classification model (non-confidential, confidential, strictly confidential). For confidential data you should preferably use **M365 Copilot Chat with your KU Leuven account**. If you use another GenAI tool, you must be **absolutely certain** it does **not store or reuse the data you enter. Try to avoid entering these data. Do so only if strictly necessary, and only to the extent required. For personal data, work with anonymous or pseudonymized data.** Additionally, in case of strictly confidential data this should first be discussed with the teaching staff of the course or your thesis supervisor.
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account. If you use another GenAI tool, you must be **absolutely certain** it does n**ot store or reuse the data you enter.**

- GenAI assistance may not be used for data or topics covered by a Non-Disclosure Agreement (NDA). Even in the absence of an NDA, certain information may still need to be treated as confidential—for example, due to regulatory requirements or the risk of significant harm to the university if disclosed. In case of doubt, check with your teaching staff or supervisor.

- If your master's thesis is under embargo, you should first discuss with your supervisor whether the use of GenAI is permitted.

- Before using a GenAI tool, always consider whether its use is responsible, including from a sustainability perspective (e.g. when using GenAI as a search engine, language assistant,...).

- **Take a scientific and critical attitude** when interacting with GenAI assistance and interpreting its output, that may not always be correct.

- As a student you are responsible for complying with Article 84 of the Regulations on Education and Examinations: your report or thesis should reflect your own knowledge, understanding and skills. Be aware that plagiarism rules also apply to (work that is the result of) the use of GenAI assistance tools.
  Exam Regulations Article 84: *"Every conduct individual students display with which they (partially) inhibit or attempt to inhibit a correct judgement of their own knowledge, understanding and/or skills or those of other students, is considered an irregularity which may result in a suitable penalty. A special type of irregularity is plagiarism, i.e. copying the work (ideas, texts, structures, designs, images, plans, codes , ...) of others or prior personal work in an exact or slightly modified way without adequately acknowledging the sources. Every possession of prohibited resources during an examination (see article 65) is considered an irregularity."*

- In order to maintain academic integrity and avoid plagiarism, **more information about being transparent on the use of GenAI assistance and about correctly citing and referencing GenAI** can be found on this website for students (Dutch/English).

- **Additional reading: KU Leuven guidelines on responsible use of Generative AI tools, and other information** (Dutch/English)

A few final words

If you are uncertain whether or not you should declare your use of GenAI tools, we suggest that you discuss this with your instructor or supervisor. It is always safer to declare GenAI use, even when it is not strictly required. However, declaring GenAI use does not entail that its use

is allowed; the right column in the table below provides more detailed instructions in this regard (code of conduct).

Moreover, advanced AI tools are evolving rapidly, and their capabilities have expanded significantly in a short period of time. As a result, we do not yet have all the answers about their responsible use. Finally, it is important to follow-up on the most recent evolutions in AI technologies, to have an open mind but also to be a bit cautious, to communicate with instructors, teaching assistants, supervisors and peers, to be as transparent as we can, and to learn together as we move along.

Student name: Anupam Chauhan **Student number:** r1018977

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As a search engine to get information on a topic or to search for existing research on the topic	*Claude Code - used to search highly specific information. Citation requested at each answer and manually verified before use.*	This use is similar to e.g. a Google search or checking Wikipedia. If you write your own text based on this information, you do not have to refer to the use of GenAI in the text. **You only have to refer to the existing research and references you have checked and used** (without such references you would be committing plagiarism and thus committing an irregularity). **However, be careful:** • Be aware that the output of the GenAI tool cannot be guaranteed as a 100% reliable source of information. The output may not be entirely correct and/or be limited due to the databases it uses. Moreover, knowledge evolves and may change over time; therefore, the database of the GenAI tool may not be up to date. Therefore, **verify the information and do not just copy-paste it** as you should understand and critically process everything you are writing.

For literature search	*Claude code - To find highly specific information as per need and provide peer-reviewed research papers, manually verified myself.*	This use is comparable to e.g. a Google Scholar search. You do not have to refer to such kind of GenAI use; **you only have to refer to the literature references you have checked and used** (without such references you would be committing plagiarism and thus committing an irregularity). **However, be careful:** • Be aware that the search output is restricted to the database the GenAI tool is built on. After this initial search, look for scientific sources and conduct your own analysis of the source documents. **Interpret, analyse and process the information you obtained; verify it and do not just copy-paste it** as you should understand and critically process everything you are writing. • Be aware that some GenAI tools may **output no or wrong references.** As a student you are responsible for further checking and verifying the absence or correctness of references; do not just copy-paste it.
For generating programming code	*Claude Code - Provided the algorithm logic myself and used to generate and debug code and to generate data processing code. Also used to improve the code written by me.*	Use of GenAI for coding is allowed except when it is prohibited by the teacher or the program of study. If used for coding, correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparant about the use of GenAI.

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Other use (specify here; this may also include a combination of types of use mentioned above):	*Claude code - used to generate figures from my hand drawn illustrations and generate good quality graphs from my existing code.*	To make sure other use of GenAI is allowed within the course or thesis, and if so, the conditions that may apply, contact the teaching staff of the course or the supervisor of the thesis **beforehand** and explain the intended GenAI use. Also inform the programme director. Motivate how you would comply with Article 84 of the exam regulations. Explain the use and the added value of the AI tool you consider to use and how it is in accordance with the assignment or thesis and the KU Leuven guidelines on responsible use of Generative AI tools. Depending on the kind of GenAI use, it may be needed to properly reference it in the text, in accordance with the instructions page on being transparent about the use of GenAI.